

THE DERIVATIVE

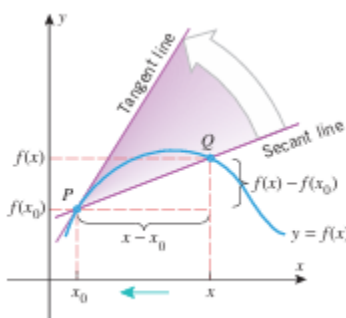
2.1.1 DEFINITION Suppose that x_0 is in the domain of the function f . The *tangent line* to the curve $y = f(x)$ at the point $P(x_0, f(x_0))$ is the line with equation

$$y - f(x_0) = m_{\text{tan}}(x - x_0)$$

where

$$m_{\text{tan}} = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \quad (1)$$

provided the limit exists. For simplicity, we will also call this the tangent line to $y = f(x)$ at x_0 .



► **Example 1** Use Definition 2.1.1 to find an equation for the tangent line to the parabola $y = x^2$ at the point $P(1, 1)$, and confirm the result agrees with that obtained in Example 1 of Section 1.1.

Solution. Applying Formula (1) with $f(x) = x^2$ and $x_0 = 1$, we have

$$\begin{aligned} m_{\text{tan}} &= \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} \\ &= \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} \\ &= \lim_{x \rightarrow 1} \frac{(x - 1)(x + 1)}{x - 1} = \lim_{x \rightarrow 1} (x + 1) = 2 \end{aligned}$$

Thus, the tangent line to $y = x^2$ at $(1, 1)$ has equation

$$y - 1 = 2(x - 1) \quad \text{or equivalently} \quad y = 2x - 1$$

which agrees with Example 1 of Section 1.1. ◀

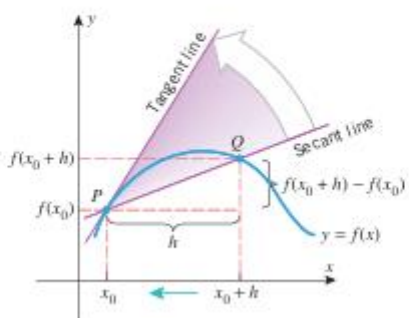
There is an alternative way of expressing Formula (1) that is commonly used. If we let h denote the difference

$$h = x - x_0$$

then the statement that $x \rightarrow x_0$ is equivalent to the statement $h \rightarrow 0$, so we can rewrite (1) in terms of x_0 and h as

$$m_{\text{tan}} = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \quad (2)$$

Figure 2.1.2 shows how Formula (2) expresses the slope of the tangent line as a limit of slopes of secant lines.



► Figure 2.1.2

► **Example 2** Compute the slope in Example 1 using Formula (2).

Solution. Applying Formula (2) with $f(x) = x^2$ and $x_0 = 1$, we obtain

$$\begin{aligned} m_{\text{tan}} &= \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(1+h)^2 - 1^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{1 + 2h + h^2 - 1}{h} = \lim_{h \rightarrow 0} (2 + h) = 2 \end{aligned}$$

which agrees with the slope found in Example 1. ◀

► **Example 3** Find an equation for the tangent line to the curve $y = 2/x$ at the point $(2, 1)$ on this curve.

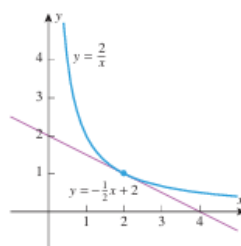
Solution. First, we will find the slope of the tangent line by applying Formula (2) with $f(x) = 2/x$ and $x_0 = 2$. This yields

$$\begin{aligned} m_{\text{tan}} &= \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{2}{2+h} - 1}{h} = \lim_{h \rightarrow 0} \frac{\left(\frac{2 - (2+h)}{2+h}\right)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-h}{h(2+h)} = - \left(\lim_{h \rightarrow 0} \frac{1}{2+h} \right) = -\frac{1}{2} \end{aligned}$$

Thus, an equation of the tangent line at $(2, 1)$ is

$$y - 1 = -\frac{1}{2}(x - 2) \quad \text{or equivalently} \quad y = -\frac{1}{2}x + 2$$

(see Figure 2.1.3). ◀



▲ Figure 2.1.3

► **Example 4** Find the slopes of the tangent lines to the curve $y = \sqrt{x}$ at $x_0 = 1$, $x_0 = 4$, and $x_0 = 9$.

Solution. We could compute each of these slopes separately, but it will be more efficient to find the slope for a general value of x_0 and then substitute the specific numerical values. Proceeding in this way we obtain

$$\begin{aligned} m_{\tan} &= \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sqrt{x_0 + h} - \sqrt{x_0}}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sqrt{x_0 + h} - \sqrt{x_0}}{h} \cdot \frac{\sqrt{x_0 + h} + \sqrt{x_0}}{\sqrt{x_0 + h} + \sqrt{x_0}} \\ &= \lim_{h \rightarrow 0} \frac{x_0 + h - x_0}{h(\sqrt{x_0 + h} + \sqrt{x_0})} \\ &= \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{x_0 + h} + \sqrt{x_0})} \\ &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{x_0 + h} + \sqrt{x_0}} = \frac{1}{2\sqrt{x_0}} \end{aligned}$$

Rationalize the numerator to help eliminate the indeterminate form of the limit.

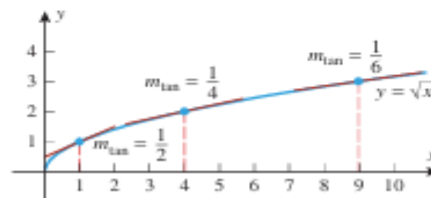
The slopes at $x_0 = 1$, 4, and 9 can now be obtained by substituting these values into our general formula for $m_{\tan}$. Thus,

$$\text{slope at } x_0 = 1: \frac{1}{2\sqrt{1}} = \frac{1}{2}$$

$$\text{slope at } x_0 = 4: \frac{1}{2\sqrt{4}} = \frac{1}{4}$$

$$\text{slope at } x_0 = 9: \frac{1}{2\sqrt{9}} = \frac{1}{6}$$

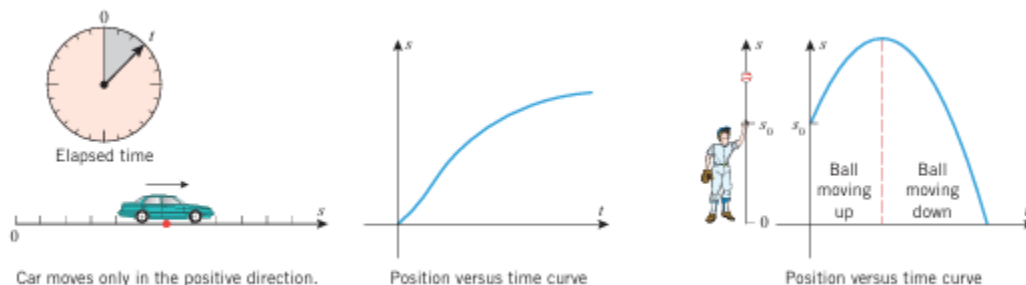
(see Figure 2.1.4). ◀



► Figure 2.1.4

VELOCITY:

One of the important themes in calculus is the study of motion. The speed and the direction of motion together comprise what is called the velocity of the object. We will only consider motion along a line; this is called rectilinear motion. For computational purposes, we will assume that a particle in rectilinear motion moves along a coordinate line, which we will call the s -axis. A graphical description of rectilinear motion along an s -axis can be obtained by making a plot of the s -coordinate of the particle versus the elapsed time t from starting time $t = 0$. This is called the position versus time curve for the particle.



▲ Figure 2.1.5

If a particle in rectilinear motion moves along an s -axis so that its position coordinate function of the elapsed time t is

$$s = f(t) \quad (3)$$

then f is called the **position function of the particle**; the graph of (3) is the position versus time curve. The **average velocity** of the particle over a time interval $[t_0, t_0 + h]$, $h > 0$, is defined to be

$$v_{\text{ave}} = \frac{\text{change in position}}{\text{time elapsed}} = \frac{f(t_0 + h) - f(t_0)}{h} \quad (4)$$

► **Example 5** Suppose that $s = f(t) = 1 + 5t - 2t^2$ is the position function of a particle, where s is in meters and t is in seconds. Find the average velocities of the particle over the time intervals (a) $[0, 2]$ and (b) $[2, 3]$.

Solution (a). Applying (4) with $t_0 = 0$ and $h = 2$, we see that the average velocity is

$$v_{\text{ave}} = \frac{f(t_0 + h) - f(t_0)}{h} = \frac{f(2) - f(0)}{2} = \frac{3 - 1}{2} = \frac{2}{2} = 1 \text{ m/s}$$

Solution (b). Applying (4) with $t_0 = 2$ and $h = 1$, we see that the average velocity is

$$v_{\text{ave}} = \frac{f(t_0 + h) - f(t_0)}{h} = \frac{f(3) - f(2)}{1} = \frac{-2 - 3}{1} = \frac{-5}{1} = -5 \text{ m/s} \quad \blacktriangleleft$$

For a particle in rectilinear motion, average velocity describes its behavior over an *interval* of time. We are interested in the particle's “instantaneous velocity,” which describes

its behavior at a specific *instant* in time. Formula (4) is not directly applicable for computing instantaneous velocity because the “time elapsed” at a specific instant is zero, so (4) is undefined. One way to circumvent this problem is to compute average velocities for small time intervals between $t = t_0$ and $t = t_0 + h$. These average velocities may be viewed as approximations to the “instantaneous velocity” of the particle at time t_0 . If these average velocities have a limit as h approaches zero, then we can take that limit to be the *instantaneous velocity* of the particle at time t_0 . Here is an example.

The change in position

$$f(t_0 + h) - f(t_0)$$

is also called the **displacement** of the particle over the time interval between t_0 and $t_0 + h$.

► **Example 6** Consider the particle in Example 5, whose position function is

$$s = f(t) = 1 + 5t - 2t^2$$

The position of the particle at time $t = 2$ s is $s = 3$ m (Figure 2.1.6). Find the particle's instantaneous velocity at time $t = 2$ s.

Solution. As a first approximation to the particle's instantaneous velocity at time $t = 2$ s, let us recall from Example 5(b) that the average velocity over the time interval from $t = 2$ to $t = 3$ is $v_{ave} = -5$ m/s. To improve on this initial approximation we will compute the average velocity over a succession of smaller and smaller time intervals. We leave it to you to verify the results in Table 2.1.1. The average velocities in this table appear to be approaching a limit of -3 m/s, providing strong evidence that the instantaneous velocity at time $t = 2$ s is -3 m/s. To confirm this analytically, we start by computing the object's average velocity over a general time interval between $t = 2$ and $t = 2 + h$ using Formula (4):

$$v_{ave} = \frac{f(2+h) - f(2)}{h} = \frac{[1 + 5(2+h) - 2(2+h)^2] - 3}{h}$$

The object's instantaneous velocity at time $t = 2$ is calculated as a limit as $h \rightarrow 0$:

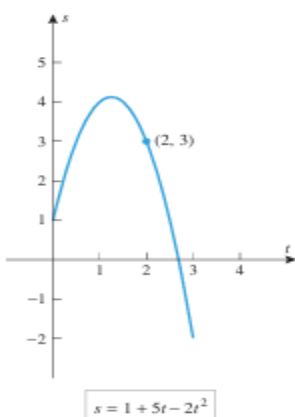
$$\begin{aligned} \text{instantaneous velocity} &= \lim_{h \rightarrow 0} \frac{[1 + 5(2+h) - 2(2+h)^2] - 3}{h} \\ &= \lim_{h \rightarrow 0} \frac{-2 + (10 + 5h) - (8 + 8h + 2h^2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-3h - 2h^2}{h} = \lim_{h \rightarrow 0} (-3 - 2h) = -3 \end{aligned}$$

This confirms our numerical conjecture that the instantaneous velocity after 2 s is -3 m/s. ◀

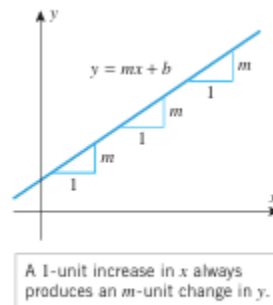
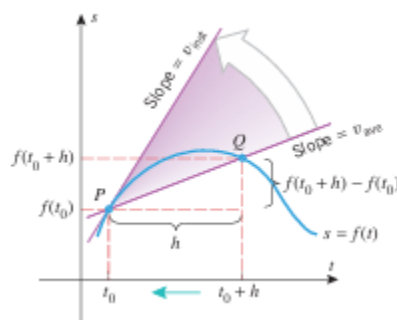
Consider a particle in rectilinear motion with position function $s = f(t)$. Motivated by Example 6, we define the instantaneous velocity v_{inst} of the particle at time t_0 to be the limit as $h \rightarrow 0$ of its average velocities v_{ave} over time intervals between $t = t_0$ and $t = t_0 + h$. Thus, from (4) we obtain

$$v_{inst} = \lim_{h \rightarrow 0} \frac{f(t_0 + h) - f(t_0)}{h} \quad (5)$$

Geometrically, the average velocity v_{ave} between $t = t_0$ and $t = t_0 + h$ is the slope of the secant line through points $P(t_0, f(t_0))$ and $Q(t_0 + h, f(t_0 + h))$ on the position versus time curve, and the instantaneous velocity v_{inst} at time t_0 is the slope of the tangent line to the position versus time curve at the point $P(t_0, f(t_0))$ (Figure 2.1.7).



▲ Figure 2.1.6



▲ Figure 2.1.8

SLOPES AND RATES OF CHANGE:

Velocity can be viewed as rate of change—the rate of change of position with respect to time.

Rates of change occur in other applications as well. Our next objective is to define precisely what is meant by the “rate of change of y with respect to x ” when y is a function of x . In the case where y is a linear function of x , say $y = mx + b$, the slope m is the natural measure of the rate of change of y with respect to x . As illustrated in Figure 2.1.8.

► **Example 7** Find the rate of change of y with respect to x if

$$(a) \ y = 2x - 1 \quad (b) \ y = -5x + 1$$

Solution. In part (a) the rate of change of y with respect to x is $m = 2$, so each 1-unit increase in x produces a 2-unit increase in y . In part (b) the rate of change of y with respect to x is $m = -5$, so each 1-unit increase in x produces a 5-unit decrease in y . ◀

In applied problems, changing the units of measurement can change the slope of a line, so it is essential to include the units when calculating the slope and describing rates of change. The following example illustrates this.

► **Example 9** Let $y = x^2 + 1$.

- (a) Find the average rate of change of y with respect to x over the interval $[3, 5]$.
- (b) Find the instantaneous rate of change of y with respect to x when $x = -4$.

Solution (a). We will apply Formula (8) with $f(x) = x^2 + 1$, $x_0 = 3$, and $x_1 = 5$. This yields

$$r_{\text{ave}} = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{f(5) - f(3)}{5 - 3} = \frac{26 - 10}{2} = 8$$

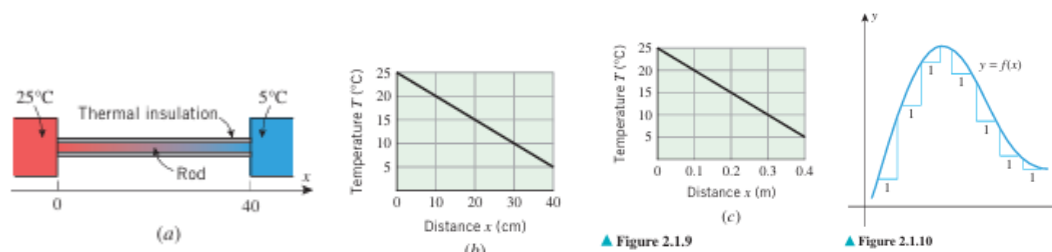
Thus, y increases an average of 8 units per unit increase in x over the interval $[3, 5]$.

Solution (b). We will apply Formula (9) with $f(x) = x^2 + 1$ and $x_0 = -4$. This yields

$$\begin{aligned} r_{\text{inst}} &= \lim_{x_1 \rightarrow x_0} \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \lim_{x_1 \rightarrow -4} \frac{f(x_1) - f(-4)}{x_1 - (-4)} = \lim_{x_1 \rightarrow -4} \frac{(x_1^2 + 1) - 17}{x_1 + 4} \\ &= \lim_{x_1 \rightarrow -4} \frac{x_1^2 - 16}{x_1 + 4} = \lim_{x_1 \rightarrow -4} \frac{(x_1 + 4)(x_1 - 4)}{x_1 + 4} = \lim_{x_1 \rightarrow -4} (x_1 - 4) = -8 \end{aligned}$$

Thus, a small increase in x from $x = -4$ will produce approximately an 8-fold decrease in y . ◀

► **Example 8** Suppose that a uniform rod of length 40 cm ($= 0.4$ m) is thermally insulated around the lateral surface and that the exposed ends of the rod are held at constant temperatures of 25°C and 5°C , respectively (Figure 2.1.9a). It is shown in physics that under appropriate conditions the graph of the temperature T versus the distance x from the left-hand end of the rod will be a straight line. Parts (b) and (c) of Figure 2.1.9 show two



such graphs: one in which x is measured in centimeters and one in which it is measured in meters. The slopes in the two cases are

$$m = \frac{5 - 25}{40 - 0} = \frac{-20}{40} = -0.5 \quad (6)$$

$$m = \frac{5 - 25}{0.4 - 0} = \frac{-20}{0.4} = -50 \quad (7)$$

The slope in (6) implies that the temperature *decreases* at a rate of 0.5°C per centimeter of distance from the left end of the rod, and the slope in (7) implies that the temperature decreases at a rate of 50°C per meter of distance from the left end of the rod. The two statements are equivalent physically, even though the slopes differ. ◀

Although the rate of change of y with respect to x is constant along a nonvertical line $y = mx + b$, this is not true for a general curve $y = f(x)$. For example, in Figure 2.1.10 the change in y that results from a 1-unit increase in x tends to have greater magnitude in regions where the curve rises or falls rapidly than in regions where it rises or falls slowly. As with velocity, we will distinguish between the average rate of change over an interval and the instantaneous rate of change at a specific point.

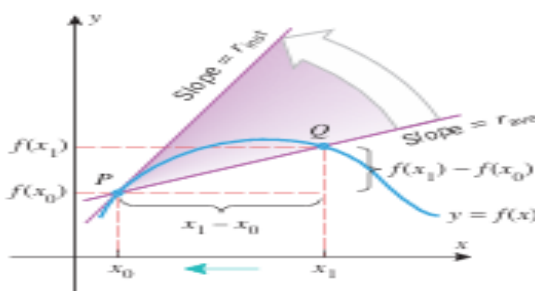
If $y = f(x)$, then we define the **average rate of change of y with respect to x over the interval $[x_0, x_1]$** to be

$$r_{\text{ave}} = \frac{f(x_1) - f(x_0)}{x_1 - x_0} \quad (8)$$

and we define the **instantaneous rate of change of y with respect to x at x_0** to be

$$r_{\text{inst}} = \lim_{x_1 \rightarrow x_0} \frac{f(x_1) - f(x_0)}{x_1 - x_0} \quad (9)$$

Geometrically, the average rate of change of y with respect to x over the interval $[x_0, x_1]$ is the slope of the secant line through the points $P(x_0, f(x_0))$ and $Q(x_1, f(x_1))$ (Figure 2.1.11), and the instantaneous rate of change of y with respect to x at x_0 is the slope of the tangent line at the point $P(x_0, f(x_0))$ (since it is the limit of the slopes of the secant lines through P).



► Figure 2.1.11

If desired, we can let $h = x_1 - x_0$, and rewrite (8) and (9) as

$$r_{\text{ave}} = \frac{f(x_0 + h) - f(x_0)}{h} \quad (10)$$

$$r_{\text{inst}} = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \quad (11)$$

2.1 TANGENT LINES AND RATES OF CHANGE:

2.2.1 DEFINITION The function f' defined by the formula

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (2)$$

is called the **derivative of f with respect to x** . The domain of f' consists of all x in the domain of f for which the limit exists.

The term “derivative” is used because the function f' is *derived* from the function f by a limiting process.

Note: it is also called the difference quotient.

► **Example 1** Find the derivative with respect to x of $f(x) = x^2$, and use it to find the equation of the tangent line to $y = x^2$ at $x = 2$.

Solution. It follows from (2) that

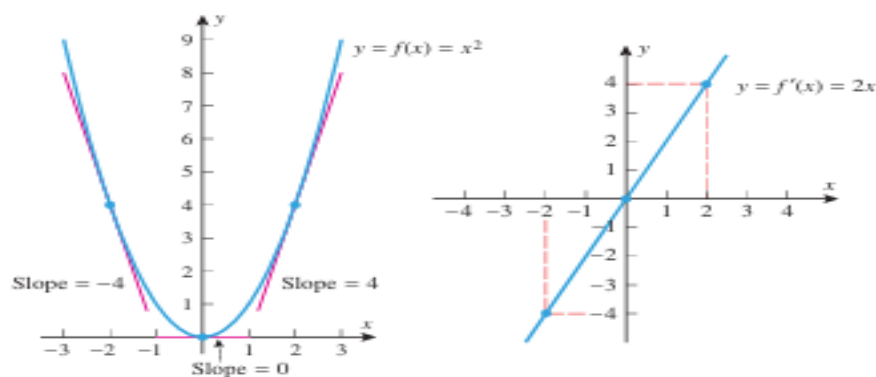
$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h} = \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} \\ &= \lim_{h \rightarrow 0} (2x + h) = 2x \end{aligned}$$

Thus, the slope of the tangent line to $y = x^2$ at $x = 2$ is $f'(2) = 4$. Since $y = 4$ if $x = 2$, the point-slope form of the tangent line is

$$y - 4 = 4(x - 2)$$

which we can rewrite in slope-intercept form as $y = 4x - 4$ (Figure 2.2.1). ◀

You can think of f' as a “slope-producing function” in the sense that the value of $f'(x)$ at $x = x_0$ is the slope of the tangent line to the graph of f at $x = x_0$. This aspect of the derivative is illustrated in Figure 2.2.2, which shows the graphs of $f(x) = x^2$ and its derivative $f'(x) = 2x$ (obtained in Example 1). The figure illustrates that the values of $f'(x) = 2x$ at $x = -2, 0$, and 2 correspond to the slopes of the tangent lines to the graph of $f(x) = x^2$ at those values of x .



In general, if $f'(x)$ is defined at $x = x_0$, then the point-slope form of the equation of the tangent line to the graph of $y = f(x)$ at $x = x_0$ may be found using the following steps.

Finding an Equation for the Tangent Line to $y = f(x)$ at $x = x_0$.

Step 1. Evaluate $f(x_0)$; the point of tangency is $(x_0, f(x_0))$.

Step 2. Find $f'(x)$ and evaluate $f'(x_0)$, which is the slope m of the line.

Step 3. Substitute the value of the slope m and the point $(x_0, f(x_0))$ into the point-slope form of the line

$$y - f(x_0) = f'(x_0)(x - x_0)$$

or, equivalently,

$$y = f(x_0) + f'(x_0)(x - x_0) \quad (3)$$

► Example 2

- (a) Find the derivative with respect to x of $f(x) = x^3 - x$.
 (b) Graph f and f' together, and discuss the relationship between the two graphs.

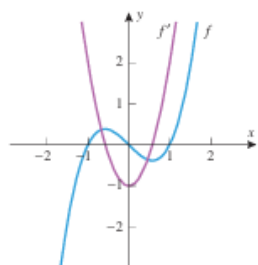
Solution (a).

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{[(x+h)^3 - (x+h)] - [x^3 - x]}{h} \\ &= \lim_{h \rightarrow 0} \frac{[x^3 + 3x^2h + 3xh^2 + h^3 - x - h] - [x^3 - x]}{h} \\ &= \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3 - h}{h} \\ &= \lim_{h \rightarrow 0} [3x^2 + 3xh + h^2 - 1] = 3x^2 - 1 \end{aligned}$$

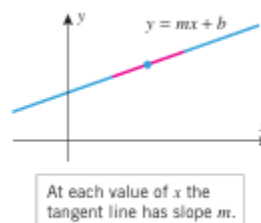
Solution (b). Since $f'(x)$ can be interpreted as the slope of the tangent line to the graph of $y = f(x)$ at x , it follows that $f'(x)$ is positive where the tangent line has positive slope, is negative where the tangent line has negative slope, and is zero where the tangent line is horizontal. We leave it for you to verify that this is consistent with the graphs of $f(x) = x^3 - x$ and $f'(x) = 3x^2 - 1$ shown in Figure 2.2.3. ◀

► Example 3 At each value of x , the tangent line to a line $y = mx + b$ coincides with the line itself (Figure 2.2.4), and hence all tangent lines have slope m . This suggests geometrically that if $f(x) = mx + b$, then $f'(x) = m$ for all x . This is confirmed by the following computations:

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{[m(x+h) + b] - [mx + b]}{h} \\ &= \lim_{h \rightarrow 0} \frac{mh}{h} = \lim_{h \rightarrow 0} m = m \quad \blacktriangleleft \end{aligned}$$



▲ Figure 2.2.3



▲ Figure 2.2.4

► **Example 4**

- (a) Find the derivative with respect to x of $f(x) = \sqrt{x}$.
 (b) Find the slope of the tangent line to $y = \sqrt{x}$ at $x = 9$.
 (c) Find the limits of $f'(x)$ as $x \rightarrow 0^+$ and as $x \rightarrow +\infty$, and explain what those limits say about the graph of f .

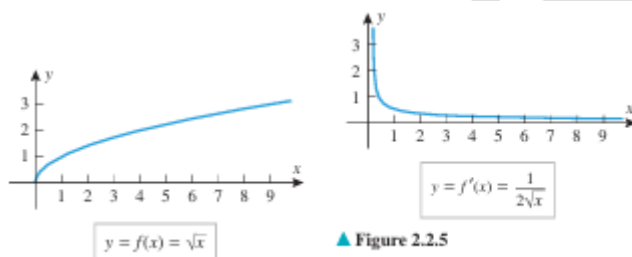
Solution (a). Recall from Example 4 of Section 2.1 that the slope of the tangent line to $y = \sqrt{x}$ at $x = x_0$ is given by $m_{\text{tan}} = 1/(2\sqrt{x_0})$. Thus, $f'(x) = 1/(2\sqrt{x})$.

Solution (b). The slope of the tangent line at $x = 9$ is $f'(9)$. From part (a), this slope is $f'(9) = 1/(2\sqrt{9}) = \frac{1}{6}$.

Solution (c). The graphs of $f(x) = \sqrt{x}$ and $f'(x) = 1/(2\sqrt{x})$ are shown in Figure 2.2.5. Observe that $f'(x) > 0$ if $x > 0$, which means that all tangent lines to the graph of $y = \sqrt{x}$ have positive slope at all points in this interval. Since

$$\lim_{x \rightarrow 0^+} \frac{1}{2\sqrt{x}} = +\infty \quad \text{and} \quad \lim_{x \rightarrow +\infty} \frac{1}{2\sqrt{x}} = 0$$

the graph of f becomes more and more vertical as $x \rightarrow 0^+$ and more and more horizontal as $x \rightarrow +\infty$. ◀



COMPUTING INSTANTANEOUS VELOCITY:

Since the right side of this equation is the derivative of the function f (with t rather than x as the independent variable), it follows that if $f(t)$ is the position function of a particle in rectilinear motion, then the function

$$v(t) = f'(t) = \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h} \quad (4)$$

represents the instantaneous velocity of the particle at time t . Accordingly, we call (4) the **instantaneous velocity function** or, more simply, the **velocity function** of the particle.

► **Example 5** Recall the particle from Example 5 of Section 2.1 with position function $s = f(t) = 1 + 5t - 2t^2$. Here $f(t)$ is measured in meters and t is measured in seconds. Find the velocity function of the particle.

Solution. It follows from (4) that the velocity function is

$$\begin{aligned} v(t) &= \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h} = \lim_{h \rightarrow 0} \frac{[1 + 5(t+h) - 2(t+h)^2] - [1 + 5t - 2t^2]}{h} \\ &= \lim_{h \rightarrow 0} \frac{-2[t^2 + 2th + h^2 - t^2] + 5h}{h} = \lim_{h \rightarrow 0} \frac{-4th - 2h^2 + 5h}{h} \\ &= \lim_{h \rightarrow 0} (-4t - 2h + 5) = 5 - 4t \end{aligned}$$

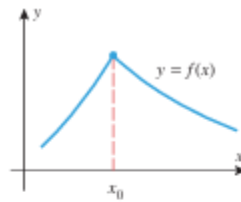
where the units of velocity are meters per second. ◀

DIFFERENTIABILITY:

2.2.2 DEFINITION A function f is said to be **differentiable at x_0** if the limit

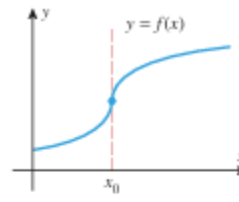
$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \quad (5)$$

exists. If f is differentiable at each point of the open interval (a, b) , then we say that it is **differentiable on (a, b)** , and similarly for open intervals of the form $(a, +\infty)$, $(-\infty, b)$, and $(-\infty, +\infty)$. In the last case we say that f is **differentiable everywhere**.

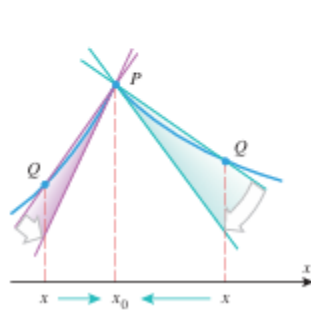


► Figure 2.2.6

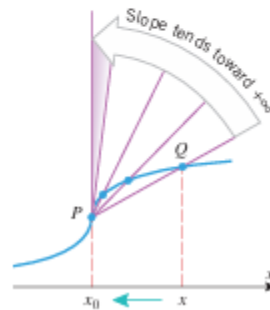
Corner point



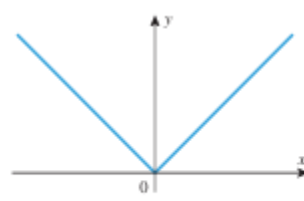
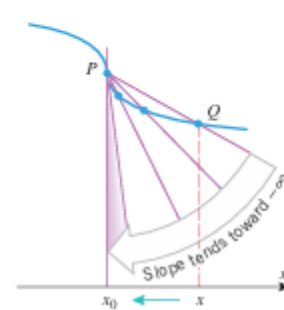
Point of vertical tangency



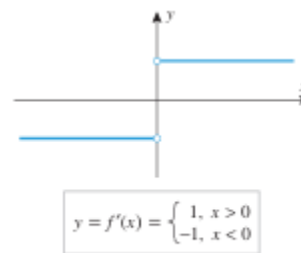
▲ Figure 2.2.7



▲ Figure 2.2.8



▲ Figure 2.2.10



▲ Figure 2.2.11

Types where function is not differentiable:

The function is given in piecewise form.

- Corner: if $f(x)$ is continuous at $x=a$, L.H.D and R.H.D exist but not equal.
- Vertical tangent: if L.H.D = R.H.D = $+\infty$ or $-\infty$.
- Vertical Cusp: if $f(x)$ is continuous at $x = a$ and L.H.D = $+\infty$ and R.H.D = $-\infty$.

► **Example 6** The graph of $y = |x|$ in Figure 2.2.10 has a corner at $x = 0$, which implies that $f(x) = |x|$ is not differentiable at $x = 0$.

- (a) Prove that $f(x) = |x|$ is not differentiable at $x = 0$ by showing that the limit in Definition 2.2.2 does not exist at $x = 0$.
 (b) Find a formula for $f'(x)$.

Solution (a). From Formula (5) with $x_0 = 0$, the value of $f'(0)$, if it were to exist, would be given by

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{|h| - |0|}{h} = \lim_{h \rightarrow 0} \frac{|h|}{h} \quad (6)$$

But

$$\frac{|h|}{h} = \begin{cases} 1, & h > 0 \\ -1, & h < 0 \end{cases}$$

so that

$$\lim_{h \rightarrow 0^-} \frac{|h|}{h} = -1 \quad \text{and} \quad \lim_{h \rightarrow 0^+} \frac{|h|}{h} = 1$$

Since these one-sided limits are not equal, the two-sided limit in (5) does not exist, and hence f is not differentiable at $x = 0$.

Solution (b). A formula for the derivative of $f(x) = |x|$ can be obtained by writing $|x|$ in piecewise form and treating the cases $x > 0$ and $x < 0$ separately. If $x > 0$, then $f(x) = x$ and $f'(x) = 1$; if $x < 0$, then $f(x) = -x$ and $f'(x) = -1$. Thus,

$$f'(x) = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}$$

The graph of f' is shown in Figure 2.2.11. Observe that f' is not continuous at $x = 0$, so this example shows that a function that is continuous everywhere may have a derivative that fails to be continuous everywhere. ◀

THE RELATIONSHIP BETWEEN DIFFERENTIABILITY AND CONTINUITY:

2.2.3 THEOREM If a function f is differentiable at x_0 , then f is continuous at x_0 .

PROOF We are given that f is differentiable at x_0 , so it follows from (5) that $f'(x_0)$ exists and is given by

$$f'(x_0) = \lim_{h \rightarrow 0} \left[\frac{f(x_0 + h) - f(x_0)}{h} \right] \quad (7)$$

To show that f is continuous at x_0 , we must show that $\lim_{x \rightarrow x_0} f(x) = f(x_0)$ or, equivalently,

$$\lim_{x \rightarrow x_0} [f(x) - f(x_0)] = 0$$

Expressing this in terms of the variable $h = x - x_0$, we must prove that

$$\lim_{h \rightarrow 0} [f(x_0 + h) - f(x_0)] = 0$$

However, this can be proved using (7) as follows:

$$\begin{aligned} \lim_{h \rightarrow 0} [f(x_0 + h) - f(x_0)] &= \lim_{h \rightarrow 0} \left[\frac{f(x_0 + h) - f(x_0)}{h} \cdot h \right] \\ &= \lim_{h \rightarrow 0} \left[\frac{f(x_0 + h) - f(x_0)}{h} \right] \cdot \lim_{h \rightarrow 0} h \\ &= f'(x_0) \cdot 0 = 0 \quad \blacksquare \end{aligned}$$

DERIVATIVES AT THE ENDPOINTS OF AN INTERVAL

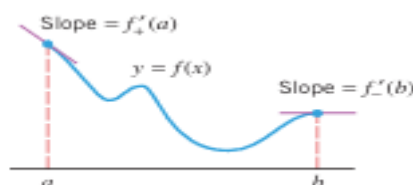
If a function f is defined on a closed interval $[a, b]$ but not outside that interval, then f' is not defined at the endpoints of the interval because derivatives are two-sided limits. To deal with this we define **left-hand derivatives** and **right-hand derivatives** by

$$f'_-(x) = \lim_{h \rightarrow 0^-} \frac{f(x+h) - f(x)}{h} \quad \text{and} \quad f'_+(x) = \lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h}$$

respectively. These are called **one-sided derivatives**. Geometrically, $f'_-(x)$ is the limit of the slopes of the secant lines as x is approached from the left and $f'_+(x)$ is the limit of the slopes of the secant lines as x is approached from the right. For a closed interval $[a, b]$, we will understand the derivative at the left endpoint to be $f'_+(a)$ and at the right endpoint to be $f'_-(b)$ (Figure 2.2.13).

In general, we will say that f is **differentiable** on an interval of the form $[a, b]$, $[a, +\infty)$, $(-\infty, b]$, $[a, b)$, or $(a, b]$ if it is differentiable at all points inside the interval and the appropriate one-sided derivative exists at each included endpoint.

It can be proved that a function f is continuous from the left at those points where the left-hand derivative exists and is continuous from the right at those points where the right-hand derivative exists.



▲ Figure 2.2.13

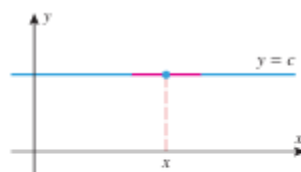
2.3 INTRODUCTION TO TECHNIQUES OF DIFFERENTIATION: DERIVATIVE OF A CONSTANT:

2.3.1 THEOREM The derivative of a constant function is 0; that is, if c is any real number, then

$$\frac{d}{dx}[c] = 0 \quad (1)$$

► Example 1

$$\frac{d}{dx}[1] = 0, \quad \frac{d}{dx}[-3] = 0, \quad \frac{d}{dx}[\pi] = 0, \quad \frac{d}{dx}[-\sqrt{2}] = 0 \quad \blacktriangleleft$$



The tangent line to the graph of $f(x) = c$ has slope 0 for all x .

▲ Figure 2.3.1

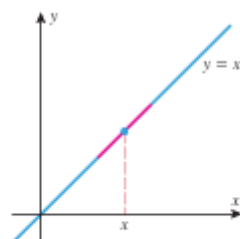
DERIVATIVES OF POWER FUNCTIONS:

2.3.2 THEOREM (The Power Rule) If n is a positive integer, then

$$\frac{d}{dx}[x^n] = nx^{n-1} \quad (5)$$

PROOF Let $f(x) = x^n$. Thus, from the definition of a derivative and the binomial formula for expanding the expression $(x + h)^n$, we obtain

$$\begin{aligned} \frac{d}{dx}[x^n] &= f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h} \\ &= \lim_{h \rightarrow 0} \frac{\left[x^n + nx^{n-1}h + \frac{n(n-1)}{2!}x^{n-2}h^2 + \cdots + nxh^{n-1} + h^n \right] - x^n}{h} \\ &= \lim_{h \rightarrow 0} \frac{nx^{n-1}h + \frac{n(n-1)}{2!}x^{n-2}h^2 + \cdots + nxh^{n-1} + h^n}{h} \\ &= \lim_{h \rightarrow 0} \left[nx^{n-1} + \frac{n(n-1)}{2!}x^{n-2}h + \cdots + nxh^{n-2} + h^{n-1} \right] \\ &= nx^{n-1} + 0 + \cdots + 0 + 0 \\ &= nx^{n-1} \quad \blacksquare \end{aligned}$$



The tangent line to the graph of $f(x) = x$ has slope 1 for all x .

▲ Figure 2.3.2

► Example 2

$$\frac{d}{dx}[x^4] = 4x^3, \quad \frac{d}{dx}[x^5] = 5x^4, \quad \frac{d}{dt}[t^{12}] = 12t^{11} \quad \blacktriangleleft$$

Although our proof of the power rule in Formula (5) applies only to *positive* integer powers of x , it is not difficult to show that the same formula holds for all integer powers of x (Exercise 82). Also, we saw in Example 4 of Section 2.2 that

$$\frac{d}{dx}[\sqrt{x}] = \frac{1}{2\sqrt{x}} \quad (6)$$

which can be expressed as

$$\frac{d}{dx}[x^{1/2}] = \frac{1}{2}x^{-1/2} = \frac{1}{2}x^{(1/2)-1}$$

2.3.3 THEOREM (Extended Power Rule) If r is any real number, then

$$\frac{d}{dx}[x^r] = rx^{r-1} \quad (7)$$

In words, to differentiate a power function, decrease the constant exponent by one and multiply the resulting power function by the original exponent.

► **Example 3**

$$\begin{aligned}\frac{d}{dx}[x^\pi] &= \pi x^{\pi-1} \\ \frac{d}{dx}\left[\frac{1}{x}\right] &= \frac{d}{dx}[x^{-1}] = (-1)x^{-1-1} = -x^{-2} = -\frac{1}{x^2} \\ \frac{d}{dw}\left[\frac{1}{w^{100}}\right] &= \frac{d}{dw}[w^{-100}] = -100w^{-101} = -\frac{100}{w^{101}} \\ \frac{d}{dx}[x^{4/5}] &= \frac{4}{5}x^{(4/5)-1} = \frac{4}{5}x^{-1/5} \\ \frac{d}{dx}[\sqrt[3]{x}] &= \frac{d}{dx}[x^{1/3}] = \frac{1}{3}x^{-2/3} = \frac{1}{3\sqrt[3]{x^2}} \quad \blacktriangleleft\end{aligned}$$

■ **DERIVATIVE OF A CONSTANT TIMES A FUNCTION**

2.3.4 THEOREM (Constant Multiple Rule) If f is differentiable at x and c is any real number, then cf is also differentiable at x and

$$\frac{d}{dx}[cf(x)] = c \frac{d}{dx}[f(x)] \quad (8)$$

PROOF

$$\begin{aligned}\frac{d}{dx}[cf(x)] &= \lim_{h \rightarrow 0} \frac{cf(x+h) - cf(x)}{h} \\ &= \lim_{h \rightarrow 0} c \left[\frac{f(x+h) - f(x)}{h} \right] \\ &= c \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= c \frac{d}{dx}[f(x)] \quad \blacksquare\end{aligned}$$

A constant factor can be moved through a limit sign.

In words, a constant factor can be moved through a derivative sign.

► **Example 4**

$$\begin{aligned}\frac{d}{dx}[4x^8] &= 4 \frac{d}{dx}[x^8] = 4[8x^7] = 32x^7 \\ \frac{d}{dx}[-x^{12}] &= (-1) \frac{d}{dx}[x^{12}] = -12x^{11} \\ \frac{d}{dx}\left[\frac{\pi}{x}\right] &= \pi \frac{d}{dx}[x^{-1}] = \pi(-x^{-2}) = -\frac{\pi}{x^2} \quad \blacktriangleleft\end{aligned}$$

DERIVATIVES OF SUMS AND DIFFERENCES

2.3.5 THEOREM (Sum and Difference Rules) If f and g are differentiable at x , then so are $f + g$ and $f - g$ and

$$\frac{d}{dx}[f(x) + g(x)] = \frac{d}{dx}[f(x)] + \frac{d}{dx}[g(x)] \quad (9)$$

$$\frac{d}{dx}[f(x) - g(x)] = \frac{d}{dx}[f(x)] - \frac{d}{dx}[g(x)] \quad (10)$$

PROOF Formula (9) can be proved as follows:

$$\begin{aligned} \frac{d}{dx}[f(x) + g(x)] &= \lim_{h \rightarrow 0} \frac{[f(x+h) + g(x+h)] - [f(x) + g(x)]}{h} \\ &= \lim_{h \rightarrow 0} \frac{[f(x+h) - f(x)] + [g(x+h) - g(x)]}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} + \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\ &= \frac{d}{dx}[f(x)] + \frac{d}{dx}[g(x)] \end{aligned}$$

The limit of a sum is the sum of the limits.

Formula (10) can be proved in a similar manner or, alternatively, by writing $f(x) - g(x)$ as $f(x) + (-1)g(x)$ and then applying Formulas (8) and (9). ■

In words, *the derivative of a sum equals the sum of the derivatives, and the derivative of a difference equals the difference of the derivatives.*

Example 5

$$\frac{d}{dx}[2x^6 + x^{-9}] = \frac{d}{dx}[2x^6] + \frac{d}{dx}[x^{-9}] = 12x^5 + (-9)x^{-10} = 12x^5 - 9x^{-10}$$

$$\begin{aligned} \frac{d}{dx}\left[\frac{\sqrt{x} - 2x}{\sqrt{x}}\right] &= \frac{d}{dx}[1 - 2\sqrt{x}] \\ &= \frac{d}{dx}[1] - \frac{d}{dx}[2\sqrt{x}] = 0 - 2\left(\frac{1}{2\sqrt{x}}\right) = -\frac{1}{\sqrt{x}} \end{aligned}$$

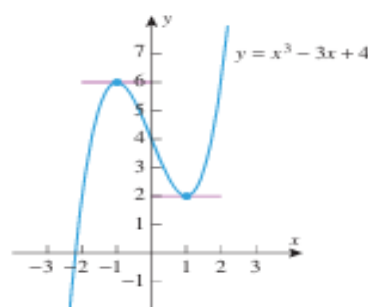
See Formula (6).

Example 7 At what points, if any, does the graph of $y = x^3 - 3x + 4$ have a horizontal tangent line?

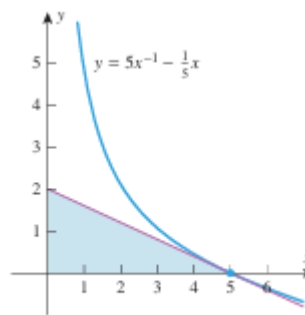
Solution. Horizontal tangent lines have slope zero, so we must find those values of x for which $y'(x) = 0$. Differentiating yields

$$y'(x) = \frac{d}{dx}[x^3 - 3x + 4] = 3x^2 - 3$$

Thus, horizontal tangent lines occur at those values of x for which $3x^2 - 3 = 0$, that is, if $x = -1$ or $x = 1$. The corresponding points on the curve $y = x^3 - 3x + 4$ are $(-1, 6)$ and $(1, 2)$ (see Figure 2.3.3). ◀



▲ Figure 2.3.3



▲ Figure 2.3.4

► **Example 8** Find the area of the triangle formed from the coordinate axes and the tangent line to the curve $y = 5x^{-1} - \frac{1}{5}x$ at the point $(5, 0)$.

Solution. Since the derivative of y with respect to x is

$$y'(x) = \frac{d}{dx} \left[5x^{-1} - \frac{1}{5}x \right] = \frac{d}{dx} [5x^{-1}] - \frac{d}{dx} \left[\frac{1}{5}x \right] = -5x^{-2} - \frac{1}{5}$$

the slope of the tangent line at the point $(5, 0)$ is $y'(5) = -\frac{2}{5}$. Thus, the equation of the tangent line at this point is

$$y - 0 = -\frac{2}{5}(x - 5) \quad \text{or equivalently} \quad y = -\frac{2}{5}x + 2$$

Since the y -intercept of this line is 2, the right triangle formed from the coordinate axes and the tangent line has legs of length 5 and 2, so its area is $\frac{1}{2}(5)(2) = 5$ (Figure 2.3.4). ◀

HIGHER DERIVATIVES

The derivative f' of a function f is itself a function and hence may have a derivative of its own. If f' is differentiable, then its derivative is denoted by f'' and is called the **second derivative** of f . As long as we have differentiability, we can continue the process

of differentiating to obtain third, fourth, fifth, and even higher derivatives of f . These successive derivatives are denoted by

$$f', \quad f'' = (f')', \quad f''' = (f'')', \quad f^{(4)} = (f''')', \quad f^{(5)} = (f^{(4)})', \dots$$

If $y = f(x)$, then successive derivatives can also be denoted by

$$y', \quad y'', \quad y''', \quad y^{(4)}, \quad y^{(5)}, \dots$$

These are called, in succession, the *first derivative*, the *second derivative*, the *third derivative*, and so forth. The number of times that f is differentiated is called the **order** of the derivative. A general n th order derivative can be denoted by

$$\frac{d^n y}{dx^n} = f^{(n)}(x) = \frac{d^n}{dx^n} [f(x)] \quad (11)$$

and the value of a general n th order derivative at a specific point $x = x_0$ can be denoted by

$$\left. \frac{d^n y}{dx^n} \right|_{x=x_0} = f^{(n)}(x_0) = \left. \frac{d^n}{dx^n} [f(x)] \right|_{x=x_0} \quad (12)$$

► **Example 9** If $f(x) = 3x^4 - 2x^3 + x^2 - 4x + 2$, then

$$f'(x) = 12x^3 - 6x^2 + 2x - 4$$

$$f''(x) = 36x^2 - 12x + 2$$

$$f'''(x) = 72x - 12$$

$$f^{(4)}(x) = 72$$

$$f^{(5)}(x) = 0$$

$$\vdots$$

$$f^{(n)}(x) = 0 \quad (n \geq 5) \quad \blacktriangleleft$$

We will discuss the significance of second derivatives and those of higher order in later sections.

2.4 THE PRODUCT AND QUOTIENT RULES:

2.4.1 THEOREM (The Product Rule) If f and g are differentiable at x , then so is the product $f \cdot g$, and

$$\frac{d}{dx}[f(x)g(x)] = f(x) \frac{d}{dx}[g(x)] + g(x) \frac{d}{dx}[f(x)] \quad (1)$$

PROOF Whereas the proofs of the derivative rules in the last section were straightforward applications of the derivative definition, a key step in this proof involves adding and subtracting the quantity $f(x+h)g(x)$ to the numerator in the derivative definition. This yields

$$\begin{aligned} \frac{d}{dx}[f(x)g(x)] &= \lim_{h \rightarrow 0} \frac{f(x+h) \cdot g(x+h) - f(x) \cdot g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x+h)g(x) + f(x+h)g(x) - f(x)g(x)}{h} \\ &= \lim_{h \rightarrow 0} \left[f(x+h) \cdot \frac{g(x+h) - g(x)}{h} + g(x) \cdot \frac{f(x+h) - f(x)}{h} \right] \\ &= \lim_{h \rightarrow 0} f(x+h) \cdot \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} + \lim_{h \rightarrow 0} g(x) \cdot \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \left[\lim_{h \rightarrow 0} f(x+h) \right] \frac{d}{dx}[g(x)] + \left[\lim_{h \rightarrow 0} g(x) \right] \frac{d}{dx}[f(x)] \\ &= f(x) \frac{d}{dx}[g(x)] + g(x) \frac{d}{dx}[f(x)] \end{aligned}$$

[Note: In the last step $f(x+h) \rightarrow f(x)$ as $h \rightarrow 0$ because f is continuous at x by Theorem 2.2.3. Also, $g(x) \rightarrow g(x)$ as $h \rightarrow 0$ because $g(x)$ does not involve h and hence is treated as constant for the limit.] ■

In words, the derivative of a product of two functions is the first function times the derivative of the second plus the second function times the derivative of the first.

► **Example 1** Find dy/dx if $y = (4x^2 - 1)(7x^3 + x)$.

Solution. There are two methods that can be used to find dy/dx . We can either use the product rule or we can multiply out the factors in y and then differentiate. We will give both methods.

Method 1. (Using the Product Rule)

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}[(4x^2 - 1)(7x^3 + x)] \\ &= (4x^2 - 1)\frac{d}{dx}[7x^3 + x] + (7x^3 + x)\frac{d}{dx}[4x^2 - 1] \\ &= (4x^2 - 1)(21x^2 + 1) + (7x^3 + x)(8x) = 140x^4 - 9x^2 - 1\end{aligned}$$

Method 2. (Multiplying First)

$$y = (4x^2 - 1)(7x^3 + x) = 28x^5 - 3x^3 - x$$

Thus,

$$\frac{dy}{dx} = \frac{d}{dx}[28x^5 - 3x^3 - x] = 140x^4 - 9x^2 - 1$$

which agrees with the result obtained using the product rule. ◀

DERIVATIVE OF A QUOTIENT:

2.4.2 THEOREM (The Quotient Rule) If f and g are both differentiable at x and if $g(x) \neq 0$, then f/g is differentiable at x and

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{g(x) \frac{d}{dx}[f(x)] - f(x) \frac{d}{dx}[g(x)]}{[g(x)]^2} \quad (2)$$

PROOF

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \lim_{h \rightarrow 0} \frac{\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)}}{h} = \lim_{h \rightarrow 0} \frac{f(x+h) \cdot g(x) - f(x) \cdot g(x+h)}{h \cdot g(x) \cdot g(x+h)}$$

Adding and subtracting $f(x) \cdot g(x)$ in the numerator yields

$$\begin{aligned}\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] &= \lim_{h \rightarrow 0} \frac{f(x+h) \cdot g(x) - f(x) \cdot g(x) - f(x) \cdot g(x+h) + f(x) \cdot g(x)}{h \cdot g(x) \cdot g(x+h)} \\ &= \lim_{h \rightarrow 0} \frac{\left[g(x) \cdot \frac{f(x+h) - f(x)}{h} \right] - \left[f(x) \cdot \frac{g(x+h) - g(x)}{h} \right]}{g(x) \cdot g(x+h)} \\ &= \frac{\lim_{h \rightarrow 0} g(x) \cdot \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} - \lim_{h \rightarrow 0} f(x) \cdot \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h}}{\lim_{h \rightarrow 0} g(x) \cdot \lim_{h \rightarrow 0} g(x+h)} \\ &= \frac{\left[\lim_{h \rightarrow 0} g(x) \right] \cdot \frac{d}{dx}[f(x)] - \left[\lim_{h \rightarrow 0} f(x) \right] \cdot \frac{d}{dx}[g(x)]}{\lim_{h \rightarrow 0} g(x) \cdot \lim_{h \rightarrow 0} g(x+h)} \\ &= \frac{g(x) \frac{d}{dx}[f(x)] - f(x) \frac{d}{dx}[g(x)]}{[g(x)]^2}\end{aligned}$$

► **Example 3** Find $y'(x)$ for $y = \frac{x^3 + 2x^2 - 1}{x + 5}$.

Solution. Applying the quotient rule yields

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} \left[\frac{x^3 + 2x^2 - 1}{x + 5} \right] = \frac{(x + 5) \frac{d}{dx} [x^3 + 2x^2 - 1] - (x^3 + 2x^2 - 1) \frac{d}{dx} [x + 5]}{(x + 5)^2} \\ &= \frac{(x + 5)(3x^2 + 4x) - (x^3 + 2x^2 - 1)(1)}{(x + 5)^2} \\ &= \frac{(3x^3 + 19x^2 + 20x) - (x^3 + 2x^2 - 1)}{(x + 5)^2} \\ &= \frac{2x^3 + 17x^2 + 20x + 1}{(x + 5)^2} \quad \blacktriangleleft\end{aligned}$$

DERIVATIVES OF TRIGONOMETRIC FUNCTIONS:

We will assume in this section that the variable x in the trigonometric functions $\sin x$, $\cos x$, $\tan x$, $\cot x$, $\sec x$, and $\csc x$ is measured in radians. Also, we will need the limits in Theorem 1.6.3, but restated as follows using h rather than x as the variable:

$$\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1 \quad \text{and} \quad \lim_{h \rightarrow 0} \frac{1 - \cos h}{h} = 0 \quad (1-2)$$

Let us start with the problem of differentiating $f(x) = \sin x$. Using the definition of the derivative we obtain

$$\begin{aligned}f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h} \quad \text{By the addition formula for sine} \\ &= \lim_{h \rightarrow 0} \left[\sin x \left(\frac{\cos h - 1}{h} \right) + \cos x \left(\frac{\sin h}{h} \right) \right] \\ &= \lim_{h \rightarrow 0} \left[\cos x \left(\frac{\sin h}{h} \right) - \sin x \left(\frac{1 - \cos h}{h} \right) \right] \quad \text{Algebraic reorganization} \\ &= \lim_{h \rightarrow 0} \cos x \cdot \lim_{h \rightarrow 0} \frac{\sin h}{h} - \lim_{h \rightarrow 0} \sin x \cdot \lim_{h \rightarrow 0} \frac{1 - \cos h}{h} \\ &= \left(\lim_{h \rightarrow 0} \cos x \right) (1) - \left(\lim_{h \rightarrow 0} \sin x \right) (0) \quad \text{Formulas (1) and (2)} \\ &= \lim_{h \rightarrow 0} \cos x = \cos x \quad \text{cos } x \text{ does not involve the variable } h \text{ and hence is treated as a constant in the limit computation.}\end{aligned}$$

Thus, we have shown that

$$\frac{d}{dx} [\sin x] = \cos x \quad (3)$$

In the exercises we will ask you to use the same method to derive the following formula for the derivative of $\cos x$:

$$\frac{d}{dx} [\cos x] = -\sin x \quad (4)$$

► **Example 1** Find dy/dx if $y = x \sin x$.

Solution. Using Formula (3) and the product rule we obtain

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}[x \sin x] \\ &= x \frac{d}{dx}[\sin x] + \sin x \frac{d}{dx}[x] \\ &= x \cos x + \sin x \quad \blacktriangleleft\end{aligned}$$

► **Example 2** Find dy/dx if $y = \frac{\sin x}{1 + \cos x}$.

Solution. Using the quotient rule together with Formulas (3) and (4) we obtain

$$\begin{aligned}\frac{dy}{dx} &= \frac{(1 + \cos x) \cdot \frac{d}{dx}[\sin x] - \sin x \cdot \frac{d}{dx}[1 + \cos x]}{(1 + \cos x)^2} \\ &= \frac{(1 + \cos x)(\cos x) - (\sin x)(-\sin x)}{(1 + \cos x)^2} \\ &= \frac{\cos x + \cos^2 x + \sin^2 x}{(1 + \cos x)^2} = \frac{\cos x + 1}{(1 + \cos x)^2} = \frac{1}{1 + \cos x} \quad \blacktriangleleft\end{aligned}$$

The derivatives of the remaining trigonometric functions are

$$\frac{d}{dx}[\tan x] = \sec^2 x \qquad \frac{d}{dx}[\sec x] = \sec x \tan x \qquad (5-6)$$

$$\frac{d}{dx}[\cot x] = -\csc^2 x \qquad \frac{d}{dx}[\csc x] = -\csc x \cot x \qquad (7-8)$$

These can all be obtained using the definition of the derivative, but it is easier to use Formulas (3) and (4) and apply the quotient rule to the relationships

$$\tan x = \frac{\sin x}{\cos x}, \quad \cot x = \frac{\cos x}{\sin x}, \quad \sec x = \frac{1}{\cos x}, \quad \csc x = \frac{1}{\sin x}$$

For example,

$$\begin{aligned}\frac{d}{dx}[\tan x] &= \frac{d}{dx} \left[\frac{\sin x}{\cos x} \right] = \frac{\cos x \cdot \frac{d}{dx}[\sin x] - \sin x \cdot \frac{d}{dx}[\cos x]}{\cos^2 x} \\ &= \frac{\cos x \cdot \cos x - \sin x \cdot (-\sin x)}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x\end{aligned}$$

► **Example 3** Find $f''(\pi/4)$ if $f(x) = \sec x$.

$$f'(x) = \sec x \tan x$$

$$\begin{aligned}f''(x) &= \sec x \cdot \frac{d}{dx}[\tan x] + \tan x \cdot \frac{d}{dx}[\sec x] \\ &= \sec x \cdot \sec^2 x + \tan x \cdot \sec x \tan x \\ &= \sec^3 x + \sec x \tan^2 x\end{aligned}$$

$$\begin{aligned}f''(\pi/4) &= \sec^3(\pi/4) + \sec(\pi/4) \tan^2(\pi/4) \\ &= (\sqrt{2})^3 + (\sqrt{2})(1)^2 = 3\sqrt{2} \quad \blacktriangleleft\end{aligned}$$

► **Example 5** As illustrated in Figure 2.5.2, suppose that a spring with an attached mass is stretched 3 cm beyond its rest position and released at time $t = 0$. Assuming that the position function of the top of the attached mass is

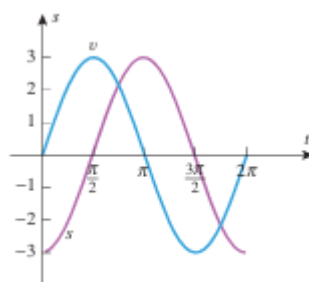
$$s = -3 \cos t \quad (10)$$

where s is in centimeters and t is in seconds, find the velocity function and discuss the motion of the attached mass.

Solution. The velocity function is

$$v = \frac{ds}{dt} = \frac{d}{dt}[-3 \cos t] = 3 \sin t$$

Figure 2.5.3 shows the graphs of the position and velocity functions. The position function tells us that the top of the mass oscillates between a low point of $s = -3$ and a high point of $s = 3$ with one complete oscillation occurring every 2π seconds [the period of (10)]. The top of the mass is moving up (the positive s -direction) when v is positive, is moving down when v is negative, and is at a high or low point when $v = 0$. Thus, for example, the top of the mass moves up from time $t = 0$ to time $t = \pi$, at which time it reaches the high point $s = 3$ and then moves down until time $t = 2\pi$, at which time it reaches the low point of $s = -3$. The motion then repeats periodically. ◀



▲ Figure 2.5.3

2.6 THE CHAIN RULE:

2.6.1 THEOREM (The Chain Rule) If g is differentiable at x and f is differentiable at $g(x)$, then the composition $f \circ g$ is differentiable at x . Moreover, if

$$y = f(g(x)) \quad \text{and} \quad u = g(x)$$

then $y = f(u)$ and

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \quad (1)$$

► **Example 1** Find dy/dx if $y = \cos(x^3)$.

Solution. Let $u = x^3$ and express y as $y = \cos u$. Applying Formula (1) yields

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} \\ &= \frac{d}{du}[\cos u] \cdot \frac{d}{dx}[x^3] \\ &= (-\sin u) \cdot (3x^2) \\ &= (-\sin(x^3)) \cdot (3x^2) = -3x^2 \sin(x^3) \quad \blacktriangleleft \end{aligned}$$

► **Example 2** Find dw/dt if $w = \tan x$ and $x = 4t^3 + t$.

Solution. In this case the chain rule computations take the form

$$\begin{aligned}\frac{dw}{dt} &= \frac{dw}{dx} \cdot \frac{dx}{dt} \\ &= \frac{d}{dx}[\tan x] \cdot \frac{d}{dt}[4t^3 + t] \\ &= (\sec^2 x) \cdot (12t^2 + 1) \\ &= [\sec^2(4t^3 + t)] \cdot (12t^2 + 1) = (12t^2 + 1) \sec^2(4t^3 + t) \quad \blacktriangleleft\end{aligned}$$

■ AN ALTERNATIVE VERSION OF THE CHAIN RULE

A convenient way to remember this formula is to call f the “outside function” and g the “inside function” in the composition $f(g(x))$ and then express (2) in words as:

The derivative of $f(g(x))$ is the derivative of the outside function evaluated at the inside function times the derivative of the inside function.

$$\frac{d}{dx}[f(g(x))] = \underbrace{f'(g(x))}_{\substack{\text{Derivative of the outside} \\ \text{function evaluated at the} \\ \text{inside function}}} \cdot \underbrace{g'(x)}_{\substack{\text{Derivative of the} \\ \text{inside function}}}$$

► **Example 3** (Example 1 revisited) Find $h'(x)$ if $h(x) = \cos(x^3)$.

Solution. We can think of h as a composition $f(g(x))$ in which $g(x) = x^3$ is the inside function and $f(x) = \cos x$ is the outside function. Thus, Formula (2) yields

$$\begin{aligned}h'(x) &= \underbrace{f'(g(x))}_{\substack{\text{Derivative of the outside} \\ \text{function evaluated at the} \\ \text{inside function}}} \cdot \underbrace{g'(x)}_{\substack{\text{Derivative of the} \\ \text{inside function}}} \\ &= f'(x^3) \cdot 3x^2 \\ &= -\sin(x^3) \cdot 3x^2 = -3x^2 \sin(x^3)\end{aligned}$$

which agrees with the result obtained in Example 1. $\blacktriangleleft$

► **Example 4**

$$\frac{d}{dx} [\tan^2 x] = \frac{d}{dx} [(\tan x)^2] = \underbrace{(2 \tan x)}_{\substack{\text{Derivative of the outside} \\ \text{function evaluated at the} \\ \text{inside function}}} \cdot \underbrace{(\sec^2 x)}_{\substack{\text{Derivative of the} \\ \text{inside function}}} = 2 \tan x \sec^2 x$$

$$\frac{d}{dx} [\sqrt{x^2 + 1}] = \underbrace{\frac{1}{2\sqrt{x^2 + 1}}}_{\substack{\text{Derivative of the outside} \\ \text{function evaluated at the} \\ \text{inside function}}} \cdot \underbrace{2x}_{\substack{\text{Derivative of the} \\ \text{inside function}}} = \frac{x}{\sqrt{x^2 + 1}} \quad \text{See Formula (6) of Section 2.3.} \quad \blacktriangleleft$$

■ **GENERALIZED DERIVATIVE FORMULAS**

There is a useful third variation of the chain rule that strikes a middle ground between Formulas (1) and (2). If we let $u = g(x)$ in (2), then we can rewrite that formula as

$$\frac{d}{dx} [f(u)] = f'(u) \frac{du}{dx} \quad (3)$$

This result, called the *generalized derivative formula* for f , provides a way of using the derivative of $f(x)$ to produce the derivative of $f(u)$, where u is a function of x . Table 2.6.1 gives some examples of this formula.

Table 2.6.1
GENERALIZED DERIVATIVE FORMULAS

$\frac{d}{dx} [u^r] = r u^{r-1} \frac{du}{dx}$	
$\frac{d}{dx} [\sin u] = \cos u \frac{du}{dx}$	$\frac{d}{dx} [\cos u] = -\sin u \frac{du}{dx}$
$\frac{d}{dx} [\tan u] = \sec^2 u \frac{du}{dx}$	$\frac{d}{dx} [\cot u] = -\csc^2 u \frac{du}{dx}$
$\frac{d}{dx} [\sec u] = \sec u \tan u \frac{du}{dx}$	$\frac{d}{dx} [\csc u] = -\csc u \cot u \frac{du}{dx}$

► **Example 5** Find

$$\begin{array}{lll} \text{(a)} \quad \frac{d}{dx}[\sin(2x)] & \text{(b)} \quad \frac{d}{dx}[\tan(x^2 + 1)] & \text{(c)} \quad \frac{d}{dx}[\sqrt{x^3 + \csc x}] \\ \text{(d)} \quad \frac{d}{dx}[x^2 - x + 2]^{3/4} & \text{(e)} \quad \frac{d}{dx}[(1 + x^5 \cot x)^{-8}] & \end{array}$$

Solution (a). Taking $u = 2x$ in the generalized derivative formula for $\sin u$ yields

$$\frac{d}{dx}[\sin(2x)] = \frac{d}{dx}[\sin u] = \cos u \frac{du}{dx} = \cos 2x \cdot \frac{d}{dx}[2x] = \cos 2x \cdot 2 = 2 \cos 2x$$

Solution (b). Taking $u = x^2 + 1$ in the generalized derivative formula for $\tan u$ yields

$$\begin{aligned} \frac{d}{dx}[\tan(x^2 + 1)] &= \frac{d}{dx}[\tan u] = \sec^2 u \frac{du}{dx} \\ &= \sec^2(x^2 + 1) \cdot \frac{d}{dx}[x^2 + 1] = \sec^2(x^2 + 1) \cdot 2x \\ &= 2x \sec^2(x^2 + 1) \end{aligned}$$

Solution (c). Taking $u = x^3 + \csc x$ in the generalized derivative formula for $\sqrt{u}$ yields

$$\begin{aligned} \frac{d}{dx}[\sqrt{x^3 + \csc x}] &= \frac{d}{dx}[\sqrt{u}] = \frac{1}{2\sqrt{u}} \frac{du}{dx} = \frac{1}{2\sqrt{x^3 + \csc x}} \cdot \frac{d}{dx}[x^3 + \csc x] \\ &= \frac{1}{2\sqrt{x^3 + \csc x}} \cdot (3x^2 - \csc x \cot x) = \frac{3x^2 - \csc x \cot x}{2\sqrt{x^3 + \csc x}} \end{aligned}$$

Solution (d). Taking $u = x^2 - x + 2$ in the generalized derivative formula for $u^{3/4}$ yields

$$\begin{aligned} \frac{d}{dx}[x^2 - x + 2]^{3/4} &= \frac{d}{dx}[u^{3/4}] = \frac{3}{4}u^{-1/4} \frac{du}{dx} \\ &= \frac{3}{4}(x^2 - x + 2)^{-1/4} \cdot \frac{d}{dx}[x^2 - x + 2] \\ &= \frac{3}{4}(x^2 - x + 2)^{-1/4}(2x - 1) \end{aligned}$$

Solution (e). Taking $u = 1 + x^5 \cot x$ in the generalized derivative formula for u^{-8} yields

$$\begin{aligned} \frac{d}{dx}[(1 + x^5 \cot x)^{-8}] &= \frac{d}{dx}[u^{-8}] = -8u^{-9} \frac{du}{dx} \\ &= -8(1 + x^5 \cot x)^{-9} \cdot \frac{d}{dx}[1 + x^5 \cot x] \\ &= -8(1 + x^5 \cot x)^{-9} \cdot [x^5(-\csc^2 x) + 5x^4 \cot x] \\ &= (8x^5 \csc^2 x - 40x^4 \cot x)(1 + x^5 \cot x)^{-9} \quad \blacktriangleleft \end{aligned}$$

Sometimes you will have to make adjustments in notation or apply the chain rule more than once to calculate a derivative.

► **Example 6** Find

$$(a) \frac{d}{dx} \left[\sin(\sqrt{1 + \cos x}) \right] \quad (b) \frac{d\mu}{dt} \text{ if } \mu = \sec \sqrt{\omega t} \quad (\omega \text{ constant})$$

Solution (a). Taking $u = \sqrt{1 + \cos x}$ in the generalized derivative formula for $\sin u$ yields

$$\begin{aligned} \frac{d}{dx} \left[\sin(\sqrt{1 + \cos x}) \right] &= \frac{d}{dx} [\sin u] = \cos u \frac{du}{dx} \\ &= \cos(\sqrt{1 + \cos x}) \cdot \frac{d}{dx} \left[\sqrt{1 + \cos x} \right] \\ &= \cos(\sqrt{1 + \cos x}) \cdot \frac{-\sin x}{2\sqrt{1 + \cos x}} \\ &= -\frac{\sin x \cos(\sqrt{1 + \cos x})}{2\sqrt{1 + \cos x}} \end{aligned}$$

We used the generalized derivative formula for $\sqrt{u}$ with $u = 1 + \cos x$.

Solution (b).

$$\begin{aligned} \frac{d\mu}{dt} &= \frac{d}{dt} [\sec \sqrt{\omega t}] = \sec \sqrt{\omega t} \tan \sqrt{\omega t} \frac{d}{dt} [\sqrt{\omega t}] \\ &= \sec \sqrt{\omega t} \tan \sqrt{\omega t} \frac{\omega}{2\sqrt{\omega t}} \end{aligned}$$

We used the generalized derivative formula for $\sec u$ with $u = \sqrt{\omega t}$.

We used the generalized derivative formula for $\sqrt{u}$ with $u = \omega t$.

FUNCTIONS DEFINED EXPLICITLY AND IMPLICITLY

An equation of the form $y = f(x)$ is said to define y **explicitly** as a function of x because the variable y appears alone on one side of the equation and does not appear at all on the other side. However, sometimes functions are defined by equations in which y is not alone on one side; for example, the equation

$$yx + y + 1 = x \quad (1)$$

is not of the form $y = f(x)$, but it still defines y as a function of x since it can be rewritten as

$$y = \frac{x - 1}{x + 1}$$

Thus, we say that (1) defines y **implicitly** as a function of x , the function being

$$f(x) = \frac{x - 1}{x + 1}$$

An equation in x and y can implicitly define more than one function of x . This can occur when the graph of the equation fails the vertical line test, so it is not the graph of a function of x . For example, if we solve the equation of the circle

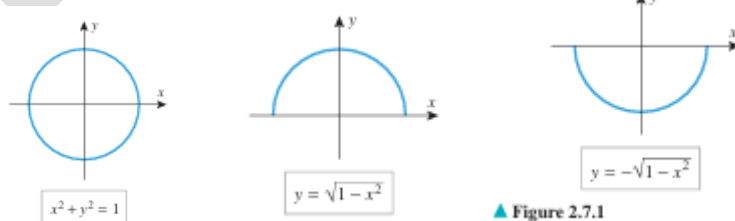
$$x^2 + y^2 = 1 \quad (2)$$

for y in terms of x , we obtain $y = \pm\sqrt{1 - x^2}$, so we have found two functions that are defined implicitly by (2), namely,

$$f_1(x) = \sqrt{1 - x^2} \quad \text{and} \quad f_2(x) = -\sqrt{1 - x^2} \quad (3)$$

The graphs of these functions are the upper and lower semicircles of the circle $x^2 + y^2 = 1$ (Figure 2.7.1). This leads us to the following definition.

2.7.1 DEFINITION We will say that a given equation in x and y defines the function f **implicitly** if the graph of $y = f(x)$ coincides with a portion of the graph of the equation.



▲ Figure 2.7.1

■ IMPLICIT DIFFERENTIATION

In general, it is not necessary to solve an equation for y in terms of x in order to differentiate the functions defined implicitly by the equation. To illustrate this, let us consider the simple equation

$$xy = 1 \quad (5)$$

One way to find dy/dx is to rewrite this equation as

$$y = \frac{1}{x} \quad (6)$$

from which it follows that

$$\frac{dy}{dx} = -\frac{1}{x^2} \quad (7)$$

Another way to obtain this derivative is to differentiate both sides of (5) *before* solving for y in terms of x , treating y as a (temporarily unspecified) differentiable function of x . With

this approach we obtain

$$\frac{d}{dx}[xy] = \frac{d}{dx}[1]$$

$$x \frac{d}{dx}[y] + y \frac{d}{dx}[x] = 0$$

$$x \frac{dy}{dx} + y = 0$$

$$\frac{dy}{dx} = -\frac{y}{x}$$

If we now substitute (6) into the last expression, we obtain

$$\frac{dy}{dx} = -\frac{1}{x^2}$$

which agrees with Equation (7). This method of obtaining derivatives is called **implicit differentiation**.

► **Example 2** Use implicit differentiation to find dy/dx if $5y^2 + \sin y = x^2$.

$$\frac{d}{dx}[5y^2 + \sin y] = \frac{d}{dx}[x^2]$$

$$5 \frac{d}{dx}[y^2] + \frac{d}{dx}[\sin y] = 2x$$

$$5 \left(2y \frac{dy}{dx} \right) + (\cos y) \frac{dy}{dx} = 2x$$

The chain rule was used here because y is a function of x .

$$10y \frac{dy}{dx} + (\cos y) \frac{dy}{dx} = 2x$$

Solving for dy/dx we obtain

$$\frac{dy}{dx} = \frac{2x}{10y + \cos y} \quad (8)$$

Note that this formula involves both x and y . In order to obtain a formula for dy/dx that involves x alone, we would have to solve the original equation for y in terms of x and then substitute in (8). However, it is impossible to do this, so we are forced to leave the formula for dy/dx in terms of x and y . ◀

► **Example 3** Use implicit differentiation to find d^2y/dx^2 if $4x^2 - 2y^2 = 9$.

Solution. Differentiating both sides of $4x^2 - 2y^2 = 9$ with respect to x yields

$$8x - 4y \frac{dy}{dx} = 0$$

from which we obtain

$$\frac{dy}{dx} = \frac{2x}{y} \quad (9)$$

Differentiating both sides of (9) yields

$$\frac{d^2y}{dx^2} = \frac{(y)(2) - (2x)(dy/dx)}{y^2} \quad (10)$$

Substituting (9) into (10) and simplifying using the original equation, we obtain

$$\frac{d^2y}{dx^2} = \frac{2y - 2x(2x/y)}{y^2} = \frac{2y^2 - 4x^2}{y^3} = -\frac{9}{y^3} \quad \blacktriangleleft$$

In Examples 2 and 3, the resulting formulas for dy/dx involved both x and y . Although it is usually more desirable to have the formula for dy/dx expressed in terms of x alone, having the formula in terms of x and y is not an impediment to finding slopes and equations of tangent lines provided the x - and y -coordinates of the point of tangency are known. This is illustrated in the following example.

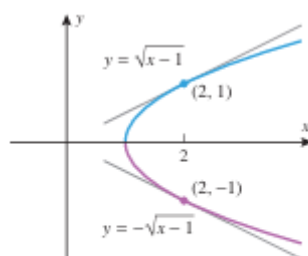
► **Example 4** Find the slopes of the tangent lines to the curve $y^2 - x + 1 = 0$ at the points $(2, -1)$ and $(2, 1)$.

Solution. We could proceed by solving the equation for y in terms of x , and then evaluating the derivative of $y = \sqrt{x-1}$ at $(2, 1)$ and the derivative of $y = -\sqrt{x-1}$ at $(2, -1)$ (Figure 2.7.4). However, implicit differentiation is more efficient since it can be used for the slopes of *both* tangent lines. Differentiating implicitly yields

$$\begin{aligned} \frac{d}{dx}[y^2 - x + 1] &= \frac{d}{dx}[0] \\ \frac{d}{dx}[y^2] - \frac{d}{dx}[x] + \frac{d}{dx}[1] &= \frac{d}{dx}[0] \\ 2y \frac{dy}{dx} - 1 &= 0 \\ \frac{dy}{dx} &= \frac{1}{2y} \end{aligned}$$

At $(2, -1)$ we have $y = -1$, and at $(2, 1)$ we have $y = 1$, so the slopes of the tangent lines to the curve at those points are

$$\left. \frac{dy}{dx} \right|_{x=2, y=-1} = -\frac{1}{2} \quad \text{and} \quad \left. \frac{dy}{dx} \right|_{x=2, y=1} = \frac{1}{2} \quad \blacktriangleleft$$



▲ Figure 2.7.4

► **Example 5**

- (a) Use implicit differentiation to find dy/dx for the Folium of Descartes $x^3 + y^3 = 3xy$.
(b) Find an equation for the tangent line to the Folium of Descartes at the point $(\frac{3}{2}, \frac{3}{2})$.
(c) At what point(s) in the first quadrant is the tangent line to the Folium of Descartes horizontal?

Solution (a). Differentiating implicitly yields

$$\begin{aligned}\frac{d}{dx}[x^3 + y^3] &= \frac{d}{dx}[3xy] \\ 3x^2 + 3y^2 \frac{dy}{dx} &= 3x \frac{dy}{dx} + 3y \\ x^2 + y^2 \frac{dy}{dx} &= x \frac{dy}{dx} + y \\ (y^2 - x) \frac{dy}{dx} &= y - x^2 \\ \frac{dy}{dx} &= \frac{y - x^2}{y^2 - x}\end{aligned}\tag{11}$$

Solution (b). At the point $(\frac{3}{2}, \frac{3}{2})$, we have $x = \frac{3}{2}$ and $y = \frac{3}{2}$, so from (11) the slope $m_{\tan}$ of the tangent line at this point is

$$m_{\tan} = \left. \frac{dy}{dx} \right|_{x=3/2, y=3/2} = \frac{(3/2) - (3/2)^2}{(3/2)^2 - (3/2)} = -1$$

Thus, the equation of the tangent line at the point $(\frac{3}{2}, \frac{3}{2})$ is

$$y - \frac{3}{2} = -1 \left(x - \frac{3}{2} \right) \quad \text{or} \quad x + y = 3$$

which is consistent with Figure 2.7.5.

Solution (c). The tangent line is horizontal at the points where $dy/dx = 0$, and from (11) this occurs only where $y - x^2 = 0$ or

$$y = x^2\tag{12}$$

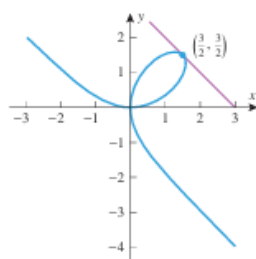
Substituting this expression for y in the equation $x^3 + y^3 = 3xy$ for the curve yields

$$\begin{aligned}x^3 + (x^2)^3 &= 3x^3 \\ x^6 - 2x^3 &= 0 \\ x^3(x^3 - 2) &= 0\end{aligned}$$

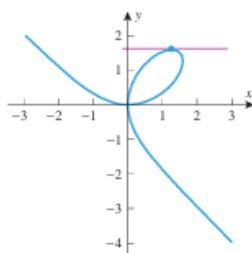
whose solutions are $x = 0$ and $x = 2^{1/3}$. From (12), the solutions $x = 0$ and $x = 2^{1/3}$ yield the points $(0, 0)$ and $(2^{1/3}, 2^{2/3})$, respectively. Of these two, only $(2^{1/3}, 2^{2/3})$ is in the first quadrant. Substituting $x = 2^{1/3}$, $y = 2^{2/3}$ into (11) yields

$$\left. \frac{dy}{dx} \right|_{x=2^{1/3}, y=2^{2/3}} = \frac{0}{2^{4/3} - 2^{2/3}} = 0$$

We conclude that $(2^{1/3}, 2^{2/3}) \approx (1.26, 1.59)$ is the only point on the Folium of Descartes in the first quadrant at which the tangent line is horizontal (Figure 2.7.6). ◀



▲ Figure 2.7.5



▲ Figure 2.7.6

DIFFERENTIABILITY OF FUNCTIONS DEFINED IMPLICITLY

When differentiating implicitly, it is assumed that y represents a differentiable function of x . If this is not so, then the resulting calculations may be nonsense. For example, if we differentiate the equation

$$x^2 + y^2 + 1 = 0 \quad (13)$$

we obtain

$$2x + 2y \frac{dy}{dx} = 0 \quad \text{or} \quad \frac{dy}{dx} = -\frac{x}{y}$$

However, this derivative is meaningless because there are no real values of x and y that satisfy (13) (why?); and hence (13) does not define any real functions implicitly.

The nonsensical conclusion of these computations conveys the importance of knowing whether an equation in x and y that is to be differentiated implicitly actually defines some differentiable function of x implicitly. Unfortunately, this can be a difficult problem, so we will leave the discussion of such matters for more advanced courses in analysis.

2.8 RELATED RATES

DIFFERENTIATING EQUATIONS TO RELATE RATES

Figure 2.8.1 shows a liquid draining through a conical filter. As the liquid drains, its volume V , height h , and radius r are functions of the elapsed time t , and at each instant these variables are related by the equation

$$V = \frac{\pi}{3} r^2 h$$

If we were interested in finding the rate of change of the volume V with respect to the time t , we could begin by differentiating both sides of this equation with respect to t to obtain

$$\frac{dV}{dt} = \frac{\pi}{3} \left[r^2 \frac{dh}{dt} + h \left(2r \frac{dr}{dt} \right) \right] = \frac{\pi}{3} \left(r^2 \frac{dh}{dt} + 2rh \frac{dr}{dt} \right)$$

Thus, to find dV/dt at a specific time t from this equation we would need to have values for r , h , dh/dt , and dr/dt at that time. This is called a **related rates problem** because the goal is to find an unknown rate of change by *relating* it to other variables whose values and whose rates of change at time t are known or can be found in some way. Let us begin with a simple example.



► **Example 1** Suppose that x and y are differentiable functions of t and are related by the equation $y = x^3$. Find dy/dt at time $t = 1$ if $x = 2$ and $dx/dt = 4$ at time $t = 1$.

Solution. Using the chain rule to differentiate both sides of the equation $y = x^3$ with respect to t yields

$$\frac{dy}{dt} = \frac{d}{dt}[x^3] = 3x^2 \frac{dx}{dt}$$

Thus, the value of dy/dt at time $t = 1$ is

$$\left. \frac{dy}{dt} \right|_{t=1} = 3(2)^2 \left. \frac{dx}{dt} \right|_{t=1} = 12 \cdot 4 = 48 \quad \blacktriangleleft$$

► **Example 2** Assume that oil spilled from a ruptured tanker spreads in a circular pattern whose radius increases at a constant rate of 2 ft/s. How fast is the area of the spill increasing when the radius of the spill is 60 ft?

Solution. Let

t = number of seconds elapsed from the time of the spill

r = radius of the spill in feet after t seconds

A = area of the spill in square feet after t seconds

(Figure 2.8.2). We know the rate at which the radius is increasing, and we want to find the rate at which the area is increasing at the instant when $r = 60$; that is, we want to find

$$\left. \frac{dA}{dt} \right|_{r=60} \quad \text{given that} \quad \frac{dr}{dt} = 2 \text{ ft/s}$$

This suggests that we look for an equation relating A and r that we can differentiate with respect to t to produce a relationship between dA/dt and dr/dt . But A is the area of a circle of radius r , so

$$A = \pi r^2 \quad (1)$$

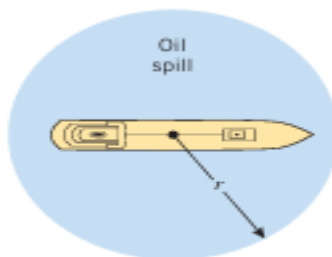
Differentiating both sides of (1) with respect to t yields

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt} \quad (2)$$

Thus, when $r = 60$ the area of the spill is increasing at the rate of

$$\left. \frac{dA}{dt} \right|_{r=60} = 2\pi(60)(2) = 240\pi \text{ ft}^2/\text{s} \approx 754 \text{ ft}^2/\text{s} \quad \blacktriangleleft$$

With some minor variations, the method used in Example 2 can be used to solve a variety of related rates problems. We can break the method down into five steps.



▲ Figure 2.8.2

A Strategy for Solving Related Rates Problems

- Step 1.** Assign letters to all quantities that vary with time and any others that seem relevant to the problem. Give a definition for each letter.
- Step 2.** Identify the rates of change that are known and the rate of change that is to be found. Interpret each rate as a derivative.
- Step 3.** Find an equation that relates the variables whose rates of change were identified in Step 2. To do this, it will often be helpful to draw an appropriately labeled figure that illustrates the relationship.
- Step 4.** Differentiate both sides of the equation obtained in Step 3 with respect to time to produce a relationship between the known rates of change and the unknown rate of change.
- Step 5.** After completing Step 4, substitute all known values for the rates of change and the variables, and then solve for the unknown rate of change.

► **Example 3** A baseball diamond is a square whose sides are 90 ft long (Figure 2.8.3). Suppose that a player running from second base to third base has a speed of 30 ft/s at the instant when he is 20 ft from third base. At what rate is the player's distance from home plate changing at that instant?

Solution. We are given a constant speed with which the player is approaching third base, and we want to find the rate of change of the distance between the player and home plate at a particular instant. Thus, let

$$\begin{aligned}t &= \text{number of seconds since the player left second base} \\x &= \text{distance in feet from the player to third base} \\y &= \text{distance in feet from the player to home plate}\end{aligned}$$

(Figure 2.8.4). Thus, we want to find

$$\left. \frac{dy}{dt} \right|_{x=20} \quad \text{given that} \quad \left. \frac{dx}{dt} \right|_{x=20} = -30 \text{ ft/s}$$

As suggested by Figure 2.8.4, an equation relating the variables x and y can be obtained using the Theorem of Pythagoras:

$$x^2 + 90^2 = y^2 \quad (3)$$

Differentiating both sides of this equation with respect to t yields

$$2x \frac{dx}{dt} = 2y \frac{dy}{dt}$$

from which we obtain

$$\frac{dy}{dt} = \frac{x}{y} \frac{dx}{dt} \quad (4)$$

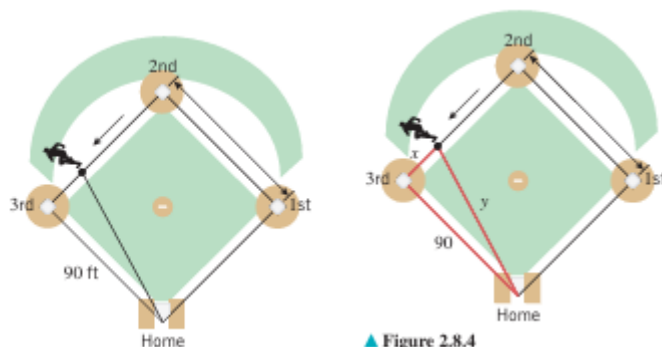
When $x = 20$, it follows from (3) that

$$y = \sqrt{20^2 + 90^2} = \sqrt{8500} = 10\sqrt{85}$$

so that (4) yields

$$\left. \frac{dy}{dt} \right|_{x=20} = \frac{20}{10\sqrt{85}}(-30) = -\frac{60}{\sqrt{85}} \approx -6.51 \text{ ft/s}$$

The negative sign in the answer tells us that y is decreasing, which makes sense physically from Figure 2.8.4. ◀



▲ Figure 2.8.4

► **Example 4** In Figure 2.8.5 we have shown a camera mounted at a point 3000 ft from the base of a rocket launching pad. If the rocket is rising vertically at 880 ft/s when it is 4000 ft above the launching pad, how fast must the camera elevation angle change at that instant to keep the camera aimed at the rocket?

Solution. Let

t = number of seconds elapsed from the time of launch

ϕ = camera elevation angle in radians after t seconds

h = height of the rocket in feet after t seconds

(Figure 2.8.6). At each instant the rate at which the camera elevation angle must change is $d\phi/dt$, and the rate at which the rocket is rising is dh/dt . We want to find

$$\left. \frac{d\phi}{dt} \right|_{h=4000} \quad \text{given that} \quad \left. \frac{dh}{dt} \right|_{h=4000} = 880 \text{ ft/s}$$

From Figure 2.8.6 we see that

$$\tan \phi = \frac{h}{3000} \quad (5)$$

Differentiating both sides of (5) with respect to t yields

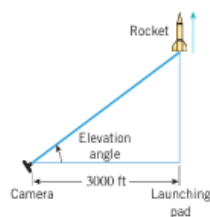
$$(\sec^2 \phi) \frac{d\phi}{dt} = \frac{1}{3000} \frac{dh}{dt} \quad (6)$$

When $h = 4000$, it follows that

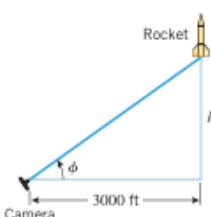
$$(\sec \phi)|_{h=4000} = \frac{5000}{3000} = \frac{5}{3}$$

(see Figure 2.8.7), so that from (6)

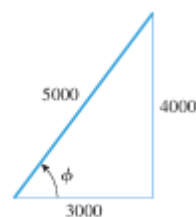
$$\begin{aligned} \left(\frac{5}{3} \right)^2 \left. \frac{d\phi}{dt} \right|_{h=4000} &= \frac{1}{3000} \cdot 880 = \frac{22}{75} \\ \left. \frac{d\phi}{dt} \right|_{h=4000} &= \frac{22}{75} \cdot \frac{9}{25} = \frac{66}{625} \approx 0.11 \text{ rad/s} \approx 6.05 \text{ deg/s} \quad \blacktriangleleft \end{aligned}$$



▲ Figure 2.8.5



▲ Figure 2.8.6



▲ Figure 2.8.7

► **Example 5** Suppose that liquid is to be cleared of sediment by allowing it to drain through a conical filter that is 16 cm high and has a radius of 4 cm at the top (Figure 2.8.8). Suppose also that the liquid is forced out of the cone at a constant rate of $2 \text{ cm}^3/\text{min}$.

- Do you think that the depth of the liquid will decrease at a constant rate? Give a verbal argument that justifies your conclusion.
- Find a formula that expresses the rate at which the depth of the liquid is changing in terms of the depth, and use that formula to determine whether your conclusion in part (a) is correct.
- At what rate is the depth of the liquid changing at the instant when the liquid in the cone is 8 cm deep?

Solution (a). For the volume of liquid to decrease by a *fixed amount*, it requires a greater decrease in depth when the cone is close to empty than when it is almost full (Figure 2.8.9). This suggests that for the volume to decrease at a constant rate, the depth must decrease at an increasing rate.

Solution (b). Let

- t = time elapsed from the initial observation (min)
- V = volume of liquid in the cone at time t (cm^3)
- y = depth of the liquid in the cone at time t (cm)
- r = radius of the liquid surface at time t (cm)

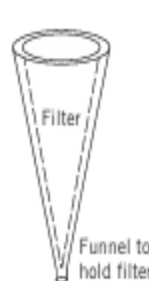
(Figure 2.8.8). At each instant the rate at which the volume of liquid is changing is dV/dt , and the rate at which the depth is changing is dy/dt . We want to express dy/dt in terms of y given that dV/dt has a constant value of $dV/dt = -2$. (We must use a minus sign here because V *decreases* as t increases.)

From the formula for the volume of a cone, the volume V , the radius r , and the depth y are related by

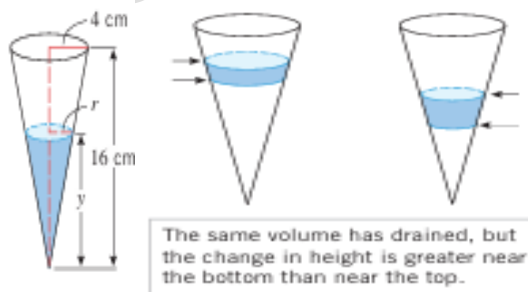
$$V = \frac{1}{3}\pi r^2 y \quad (7)$$

If we differentiate both sides of (7) with respect to t , the right side will involve the quantity dr/dt . Since we have no direct information about dr/dt , it is desirable to eliminate r from (7) before differentiating. This can be done using similar triangles. From Figure 2.8.8 we see that

$$\frac{r}{y} = \frac{4}{16} \quad \text{or} \quad r = \frac{1}{4}y$$



▲ Figure 2.8.8



▲ Figure 2.8.9

Substituting this expression in (7) gives

$$V = \frac{\pi}{48}y^3 \quad (8)$$

Differentiating both sides of (8) with respect to t we obtain

$$\frac{dV}{dt} = \frac{\pi}{48} \left(3y^2 \frac{dy}{dt} \right)$$

or

$$\frac{dy}{dt} = \frac{16}{\pi y^2} \frac{dV}{dt} = \frac{16}{\pi y^2} (-2) = -\frac{32}{\pi y^2} \quad (9)$$

which expresses dy/dt in terms of y . The minus sign tells us that y is decreasing with time, and

$$\left| \frac{dy}{dt} \right| = \frac{32}{\pi y^2}$$

tells us how fast y is decreasing. From this formula we see that $|dy/dt|$ increases as y decreases, which confirms our conjecture in part (a) that the depth of the liquid decreases more quickly as the liquid drains through the filter.

Solution (c). The rate at which the depth is changing when the depth is 8 cm can be obtained from (9) with $y = 8$:

$$\left. \frac{dy}{dt} \right|_{y=8} = -\frac{32}{\pi(8^2)} = -\frac{1}{2\pi} \approx -0.16 \text{ cm/min} \quad \blacktriangleleft$$

2.9 LOCAL LINEAR APPROXIMATION; DIFFERENTIALS

A function that is differentiable at x_0 is sometimes said to be locally linear at x_0 .

The line that best approximates the graph of f in the vicinity of $P(x_0, f(x_0))$ is the tangent line to the graph of f at x_0 , given by the equation

$$y = f(x_0) + f'(x_0)(x - x_0)$$

[see Formula (3) of Section 2.2]. Thus, for values of x near x_0 we can approximate values of $f(x)$ by

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0) \quad (1)$$

This is called the **local linear approximation** of f at x_0 . This formula can also be expressed in terms of the increment $\Delta x = x - x_0$ as

$$f(x_0 + \Delta x) \approx f(x_0) + f'(x_0)\Delta x \quad (2)$$

► Example 1

- (a) Find the local linear approximation of $f(x) = \sqrt{x}$ at $x_0 = 1$.
(b) Use the local linear approximation obtained in part (a) to approximate $\sqrt{1.1}$, and compare your approximation to the result produced directly by a calculating utility.

Solution (a). Since $f'(x) = 1/(2\sqrt{x})$, it follows from (1) that the local linear approximation of $\sqrt{x}$ at a point x_0 is

$$\sqrt{x} \approx \sqrt{x_0} + \frac{1}{2\sqrt{x_0}}(x - x_0)$$

Thus, the local linear approximation at $x_0 = 1$ is

$$\sqrt{x} \approx 1 + \frac{1}{2}(x - 1) \quad (3)$$

The graphs of $y = \sqrt{x}$ and the local linear approximation $y = 1 + \frac{1}{2}(x - 1)$ are shown in Figure 2.9.2.



▲ Figure 2.9.2

► Example 2

- (a) Find the local linear approximation of $f(x) = \sin x$ at $x_0 = 0$.
(b) Use the local linear approximation obtained in part (a) to approximate $\sin 2^\circ$, and compare your approximation to the result produced directly by your calculating device.

Solution (a). Since $f'(x) = \cos x$, it follows from (1) that the local linear approximation of $\sin x$ at a point x_0 is

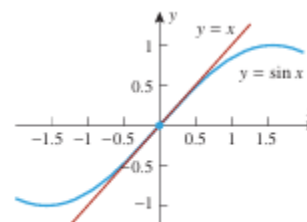
$$\sin x \approx \sin x_0 + (\cos x_0)(x - x_0)$$

Thus, the local linear approximation at $x_0 = 0$ is

$$\sin x \approx \sin 0 + (\cos 0)(x - 0)$$

which simplifies to

$$\sin x \approx x \quad (4)$$



▲ Figure 2.9.3

Solution (b). The variable x in (4) is in radian measure, so we must first convert 2° to radians before we can apply this approximation. Since

$$2^\circ = 2 \left(\frac{\pi}{180} \right) = \frac{\pi}{90} \approx 0.0349066 \text{ radian}$$

it follows from (4) that $\sin 2^\circ \approx 0.0349066$. Comparing the two graphs in Figure 2.9.3, we would expect this approximation to be slightly larger than the exact value. The calculator approximation $\sin 2^\circ \approx 0.0348995$ shows that this is indeed the case. ◀

■ DIFFERENTIALS

The Leibniz notation dy/dx eventually prevailed because it suggests correct formulas in a natural way, the chain rule

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

being a good example. Up to now we have interpreted dy/dx as a single entity representing the derivative of y with respect to x ; the symbols “ dy ” and “ dx ,” which are called differentials.

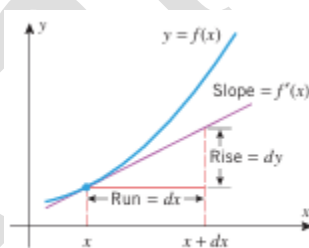
Assume that f is differentiable at a point x , define dx to be an independent variable that can have any real value, and define dy by the formula

$$dy = f'(x) dx \quad (5)$$

If $dx \neq 0$, then we can divide both sides of (5) by dx to obtain

$$\frac{dy}{dx} = f'(x) \quad (6)$$

Thus, we have achieved our goal of defining dy and dx so their ratio is $f'(x)$. Formula (5) is said to express (6) in differential form. To interpret (5) geometrically, note that $f'(x)$ is the slope of the tangent line to the graph of f at x . The differentials dy and dx can be viewed as a corresponding rise and run of this tangent line (Figure 2.9.5)



▲ Figure 2.9.5

► **Example 3** Express the derivative with respect to x of $y = x^2$ in differential form, and discuss the relationship between dy and dx at $x = 1$.

Solution. The derivative of y with respect to x is $dy/dx = 2x$, which can be expressed in differential form as

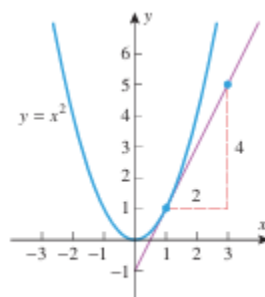
$$dy = 2x dx$$

When $x = 1$ this becomes

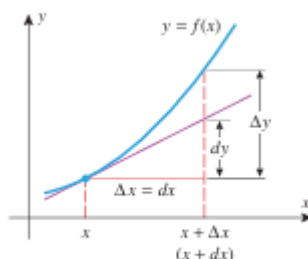
$$dy = 2 dx$$

This tells us that if we travel along the tangent line to the curve $y = x^2$ at $x = 1$, then a change of dx units in x produces a change of $2 dx$ units in y . Thus, for example, a run of $dx = 2$ units produces a rise of $dy = 4$ units along the tangent line (Figure 2.9.6). ◀

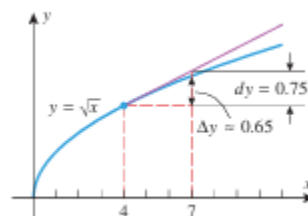
It is important to understand the distinction between the increment Δy and the differential dy . To see the difference, let us assign the independent variables dx and Δx the same value, so $dx = \Delta x$. Then Δy represents the change in y that occurs when we start at x and travel along the curve $y = f(x)$ until we have moved $\Delta x (= dx)$ units in the x -direction, while dy represents the change in y that occurs if we start at x and travel along the tangent line until we have moved $dx (= \Delta x)$ units in the x -direction (Figure 2.9.7).



▲ Figure 2.9.6



▲ Figure 2.9.7



▲ Figure 2.9.8

► **Example 4** Let $y = \sqrt{x}$.

- (a) Find formulas for Δy and dy .
 (b) Evaluate Δy and dy at $x = 4$ with $dx = \Delta x = 3$. Then make a sketch of $y = \sqrt{x}$ showing the values of Δy and dy in the picture.

Solution (a). With $y = f(x) = \sqrt{x}$ we obtain

$$\Delta y = f(x + \Delta x) - f(x) = \sqrt{x + \Delta x} - \sqrt{x}$$

and

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x}}, \quad \text{so} \quad dy = \frac{1}{2\sqrt{x}} dx$$

Solution (b). At $x = 4$ with $dx = \Delta x = 3$,

$$\Delta y = \sqrt{7} - \sqrt{4} \approx 0.65$$

and

$$dy = \frac{1}{2\sqrt{4}}(3) = \frac{3}{4} = 0.75$$

Figure 2.9.8 shows the curve $y = \sqrt{x}$ together with Δy and dy . ◀

■ LOCAL LINEAR APPROXIMATION FROM THE DIFFERENTIAL POINT OF VIEW

Although Δy and dy are generally different, the differential dy will nonetheless be a good approximation of Δy provided $dx = \Delta x$ is close to 0. To see this, recall from Section 2.2 that

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

It follows that if Δx is close to 0, then we will have $f'(x) \approx \Delta y / \Delta x$ or, equivalently,

$$\Delta y \approx f'(x) \Delta x$$

If we agree to let $dx = \Delta x$, then we can rewrite this as

$$\Delta y \approx f'(x) dx = dy \quad (7)$$

In words, this states that for values of dx near zero the differential dy closely approximates the increment Δy (Figure 2.9.7). But this is to be expected since the graph of the tangent line at x is the local linear approximation of the graph of f .

■ ERROR PROPAGATION

It refers to the process of determining how uncertainties in individual measurements affect the uncertainty of a final result calculated from those measurements. For this purpose, suppose:

x_0 is the exact value of the quantity being measured
 $y_0 = f(x_0)$ is the exact value of the quantity being computed
 x is the measured value of x_0
 $y = f(x)$ is the computed value of y

We define

$dx (= \Delta x) = x - x_0$ to be the **measurement error** of x
 $\Delta y = f(x) - f(x_0)$ to be the **propagated error** of y

It follows from (7) with x_0 replacing x that the propagated error Δy can be approximated by

$$\Delta y \approx dy = f'(x_0) dx \quad (8)$$

Unfortunately, there is a practical difficulty in applying this formula since the value of x_0 is unknown. (Keep in mind that only the measured value x is known to the researcher.) This being the case, it is standard practice in research to use the measured value x in place of x_0 in (8) and use the approximation

$$\Delta y \approx dy = f'(x) dx \quad (9)$$

for the propagated error.

► **Example 5** Suppose that the side of a square is measured with a ruler to be 10 inches with a measurement error of at most $\pm \frac{1}{32}$ in. Estimate the error in the computed area of the square.

Solution. Let x denote the exact length of a side and y the exact area so that $y = x^2$. It follows from (9) with $f(x) = x^2$ that if dx is the measurement error, then the propagated error Δy can be approximated as

$$\Delta y \approx dy = 2x dx$$

Substituting the measured value $x = 10$ into this equation yields

$$dy = 20 dx \quad (10)$$

But to say that the measurement error is at most $\pm \frac{1}{32}$ means that

$$-\frac{1}{32} \leq dx \leq \frac{1}{32}$$

Multiplying these inequalities through by 20 and applying (10) yields

$$20 \left(-\frac{1}{32}\right) \leq dy \leq 20 \left(\frac{1}{32}\right) \quad \text{or equivalently} \quad -\frac{5}{8} \leq dy \leq \frac{5}{8}$$

Thus, the propagated error in the area is estimated to be within $\pm \frac{5}{8}$ in². ◀

If the true value of a quantity is q and a measurement or calculation produces an error Δq , then $\Delta q/q$ is called the **relative error** in the measurement or calculation; when expressed

as a percentage, $\Delta q/q$ is called the **percentage error**. As a practical matter, the true value q is usually unknown, so that the measured or calculated value of q is used instead; and the relative error is approximated by dq/q .

► **Example 6** The diameter of a polyurethane sphere is measured with percentage error within $\pm 0.4\%$. Estimate the percentage error in the calculated volume of the sphere.

Solution. A sphere of diameter x has radius $r = x/2$ and volume

$$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi \left(\frac{x}{2}\right)^3 = \frac{1}{6}\pi x^3$$

Therefore

$$\frac{dV}{dx} = \frac{1}{2}\pi x^2 \quad \text{and} \quad dV = \frac{1}{2}\pi x^2 dx$$

The relative error in V is then approximately

$$\frac{dV}{V} = \frac{\frac{1}{2}\pi x^2 dx}{\frac{1}{6}\pi x^3} = 3 \frac{dx}{x} \quad (11)$$

We are given that the percentage error in the measured value of x is within $\pm 0.4\%$, which means that

$$-0.004 \leq \frac{dx}{x} \leq 0.004$$

Multiplying these inequalities through by 3 and applying (11) yields

$$-0.012 \leq \frac{dV}{V} \leq 0.012$$

Thus, we estimate the percentage error in the calculated value of V to be within $\pm 1.2\%$.

MORE NOTATION; DIFFERENTIAL FORMULAS

The symbol df is another common notation for the differential of a function $y = f(x)$. For example, if $f(x) = \sin x$, then we can write $df = \cos x dx$. We can also view the symbol “ d ” as an *operator* that acts on a function to produce the corresponding differential. For example, $d[x^2] = 2x dx$, $d[\sin x] = \cos x dx$, and so on. All of the general rules of differentiation then have corresponding differential versions:

DERIVATIVE FORMULA	DIFFERENTIAL FORMULA
$\frac{d}{dx}[c] = 0$	$d[c] = 0$
$\frac{d}{dx}[cf] = c \frac{df}{dx}$	$d[cf] = c df$
$\frac{d}{dx}[f + g] = \frac{df}{dx} + \frac{dg}{dx}$	$d[f + g] = df + dg$
$\frac{d}{dx}[fg] = f \frac{dg}{dx} + g \frac{df}{dx}$	$d[fg] = f dg + g df$
$\frac{d}{dx}\left[\frac{f}{g}\right] = \frac{g \frac{df}{dx} - f \frac{dg}{dx}}{g^2}$	$d\left[\frac{f}{g}\right] = \frac{g df - f dg}{g^2}$

For example,

$$\begin{aligned} d[x^2 \sin x] &= (x^2 \cos x + 2x \sin x) dx \\ &= x^2(\cos x dx) + (2x dx) \sin x \\ &= x^2 d[\sin x] + (\sin x) d[x^2] \end{aligned}$$

illustrates the differential version of the product rule.

Exercise 2.1

11–14 A function $y = f(x)$ and values of x_0 and x_1 are given.

- (a) Find the average rate of change of y with respect to x over the interval $[x_0, x_1]$.
- (b) Find the instantaneous rate of change of y with respect to x at the specified value of x_0 .
- (c) Find the instantaneous rate of change of y with respect to x at an arbitrary value of x_0 .
- (d) The average rate of change in part (a) is the slope of a certain secant line, and the instantaneous rate of change in part (b) is the slope of a certain tangent line. Sketch the graph of $y = f(x)$ together with those two lines. ■

11. $y = 2x^2$; $x_0 = 0$, $x_1 = 1$ **12.** $y = x^3$; $x_0 = 1$, $x_1 = 2$

13. $y = 1/x$; $x_0 = 2$, $x_1 = 3$ **14.** $y = 1/x^2$; $x_0 = 1$, $x_1 = 2$

15–18 A function $y = f(x)$ and an x -value x_0 are given.

- (a) Find a formula for the slope of the tangent line to the graph of f at a general point $x = x_0$.
- (b) Use the formula obtained in part (a) to find the slope of the tangent line for the given value of x_0 . ■

15. $f(x) = x^2 - 1$; $x_0 = -1$

16. $f(x) = x^2 + 3x + 2$; $x_0 = 2$

17. $f(x) = x + \sqrt{x}$; $x_0 = 1$

18. $f(x) = 1/\sqrt{x}$; $x_0 = 4$

Exercise 2.2

- 3.** (a) If you are given an equation for the tangent line at the point $(a, f(a))$ on a curve $y = f(x)$, how would you go about finding $f'(a)$?
- (b) Given that the tangent line to the graph of $y = f(x)$ at the point $(2, 5)$ has the equation $y = 3x - 1$, find $f'(2)$.
- (c) For the function $y = f(x)$ in part (b), what is the instantaneous rate of change of y with respect to x at $x = 2$?

9–14 Use Definition 2.2.1 to find $f'(x)$, and then find the tangent line to the graph of $y = f(x)$ at $x = a$. ■

9. $f(x) = 2x^2$; $a = 1$ **10.** $f(x) = 1/x^2$; $a = -1$

11. $f(x) = x^3$; $a = 0$ **12.** $f(x) = 2x^3 + 1$; $a = -1$

13. $f(x) = \sqrt{x+1}$; $a = 8$ **14.** $f(x) = \sqrt{2x+1}$; $a = 4$

31–32 The given limit represents $f'(a)$ for some function f and some number a . Find $f(x)$ and a in each case. ■

31. (a) $\lim_{\Delta x \rightarrow 0} \frac{\sqrt{1+\Delta x} - 1}{\Delta x}$ (b) $\lim_{x_1 \rightarrow 3} \frac{x_1^2 - 9}{x_1 - 3}$

32. (a) $\lim_{h \rightarrow 0} \frac{\cos(\pi+h) + 1}{h}$ (b) $\lim_{x \rightarrow 1} \frac{x^7 - 1}{x - 1}$

46. Show that $f(x)$ is continuous but not differentiable at the indicated point. Sketch the graph of f .

(a) $f(x) = \sqrt[3]{x}, x = 0$

(b) $f(x) = \sqrt[3]{(x-2)^2}, x = 2$

47. Show that

$$f(x) = \begin{cases} x^2 + 1, & x \leq 1 \\ 2x, & x > 1 \end{cases}$$

is continuous and differentiable at $x = 1$. Sketch the graph of f .

48. Show that

$$f(x) = \begin{cases} x^2 + 2, & x \leq 1 \\ x + 2, & x > 1 \end{cases}$$

is continuous but not differentiable at $x = 1$. Sketch the graph of f .

49. Show that

$$f(x) = \begin{cases} x \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

is continuous but not differentiable at $x = 0$. Sketch the graph of f near $x = 0$. (See Figure 1.6.6 and the remark following Example 3 in Section 1.6.)

50. Show that

$$f(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

is continuous and differentiable at $x = 0$. Sketch the graph of f near $x = 0$.

Exercise 2.3

9–16 Find $f'(x)$. ■

9. $f(x) = x^{-3} + \frac{1}{x^7}$

10. $f(x) = \sqrt{x} + \frac{1}{x}$

11. $f(x) = -3x^{-8} + 2\sqrt{x}$

12. $f(x) = 7x^{-6} - 5\sqrt{x}$

13. $f(x) = x^\pi + \frac{1}{x\sqrt{10}}$

14. $f(x) = \sqrt[3]{\frac{8}{x}}$

15. $f(x) = (3x^2 + 1)^2$

16. $f(x) = ax^3 + bx^2 + cx + d$ (a, b, c, d constant)

17–18 Find $y'(1)$. ■

17. $y = 5x^2 - 3x + 1$

18. $y = \frac{x^{3/2} + 2}{x}$

19–20 Find dx/dt . ■

19. $x = t^2 - t$

20. $x = \frac{t^2 + 1}{3t}$

41–42 Find d^2y/dx^2 . ■

41. (a) $y = 7x^3 - 5x^2 + x$

(b) $y = 12x^2 - 2x + 3$

(c) $y = \frac{x+1}{x}$

(d) $y = (5x^2 - 3)(7x^3 + x)$

42. (a) $y = 4x^7 - 5x^3 + 2x$

(b) $y = 3x + 2$

(c) $y = \frac{3x-2}{5x}$

(d) $y = (x^3 - 5)(2x + 3)$

46. Find
 (a) $y'''(0)$, where $y = 4x^4 + 2x^3 + 3$
 (b) $\left. \frac{d^4 y}{dx^4} \right|_{x=1}$, where $y = \frac{6}{x^4}$.
47. Show that $y = x^3 + 3x + 1$ satisfies $y''' + xy'' - 2y' = 0$.
48. Show that if $x \neq 0$, then $y = 1/x$ satisfies the equation $x^3 y'' + x^2 y' - xy = 0$.

Exercise 2.4

Compute the following by product and quotient Formulae:

5–20 Find $f'(x)$. ■

5. $f(x) = (3x^2 + 6)(2x - \frac{1}{4})$
6. $f(x) = (2 - x - 3x^3)(7 + x^5)$
7. $f(x) = (x^3 + 7x^2 - 8)(2x^{-3} + x^{-4})$
8. $f(x) = \left(\frac{1}{x} + \frac{1}{x^2}\right)(3x^3 + 27)$
9. $f(x) = (x - 2)(x^2 + 2x + 4)$
10. $f(x) = (x^2 + x)(x^2 - x)$
11. $f(x) = \frac{3x + 4}{x^2 + 1}$
12. $f(x) = \frac{x - 2}{x^4 + x + 1}$
13. $f(x) = \frac{x^2}{3x - 4}$
14. $f(x) = \frac{2x^2 + 5}{3x - 4}$
15. $f(x) = \frac{(2\sqrt{x} + 1)(x - 1)}{x + 3}$

31–36 Find all values of x at which the tangent line to the given curve satisfies the stated property. ■

31. $y = \frac{x^2 - 1}{x + 2}$; horizontal
32. $y = \frac{x^2 + 1}{x - 1}$; horizontal
33. $y = \frac{x^2 + 1}{x + 1}$; parallel to the line $y = x$
34. $y = \frac{x + 3}{x + 2}$; perpendicular to the line $y = x$
35. $y = \frac{1}{x + 4}$; passes through the origin
36. $y = \frac{2x + 5}{x + 2}$; y-intercept 2

Exercise 2.5

27. (a) Show that $y = x \sin x$ is a solution to $y'' + y = 2 \cos x$.
 (b) Show that $y = x \sin x$ is a solution of the equation $y^{(4)} + y'' = -2 \cos x$.
28. (a) Show that $y = \cos x$ and $y = \sin x$ are solutions of the equation $y'' + y = 0$.
 (b) Show that $y = A \sin x + B \cos x$ is a solution of the equation $y'' + y = 0$ for all constants A and B .

43. In each part, determine where f is differentiable.

- (a) $f(x) = \sin x$ (b) $f(x) = \cos x$
 (c) $f(x) = \tan x$ (d) $f(x) = \cot x$
 (e) $f(x) = \sec x$ (f) $f(x) = \csc x$
 (g) $f(x) = \frac{1}{1 + \cos x}$ (h) $f(x) = \frac{1}{\sin x \cos x}$
 (i) $f(x) = \frac{\cos x}{2 - \sin x}$

Exercise 2.6

27–40 Find dy/dx . ■

27. $y = x^3 \sin^2(5x)$ 28. $y = \sqrt{x} \tan^3(\sqrt{x})$
 29. $y = x^5 \sec(1/x)$ 30. $y = \frac{\sin x}{\sec(3x + 1)}$
 31. $y = \cos(\cos x)$ 32. $y = \sin(\tan 3x)$
 33. $y = \cos^3(\sin 2x)$ 34. $y = \frac{1 + \csc(x^2)}{1 - \cot(x^2)}$
 35. $y = (5x + 8)^7 (1 - \sqrt{x})^6$ 36. $y = (x^2 + x)^5 \sin^8 x$
 37. $y = \left(\frac{x-5}{2x+1}\right)^3$ 38. $y = \left(\frac{1+x^2}{1-x^2}\right)^{17}$
 39. $y = \frac{(2x+3)^3}{(4x^2-1)^8}$ 40. $y = [1 + \sin^3(x^5)]^{12}$

Exercise 2.7

1–2

- (a) Find dy/dx by differentiating implicitly.
 (b) Solve the equation for y as a function of x , and find dy/dx from that equation.
 (c) Confirm that the two results are consistent by expressing the derivative in part (a) as a function of x alone. ■

1. $x + xy - 2x^3 = 2$ 2. $\sqrt{y} - \sin x = 2$

22. The function

$$f(x) = \begin{cases} \sqrt{1-x^2}, & 0 < x \leq 1 \\ -\sqrt{1-x^2}, & -1 \leq x < 0 \end{cases}$$

is defined implicitly by the equation $x^2 + y^2 = 1$.

23. The function $|x|$ is not defined implicitly by the equation $(x+y)(x-y) = 0$.

24. If y is defined implicitly as a function of x by the equation $x^2 + y^2 = 1$, then $dy/dx = -x/y$.

31–34 Use implicit differentiation to find the specified derivative. ■

31. $a^4 - t^4 = 6a^2t$; da/dt 32. $\sqrt{u} + \sqrt{v} = 5$; du/dv
 33. $a^2\omega^2 + b^2\lambda^2 = 1$ (a, b constants); $d\omega/d\lambda$
 34. $y = \sin x$; dx/dy

36. (a) A student claims that the ellipse $x^2 - xy + y^2 = 1$ has a horizontal tangent line at the point $(1, 1)$. Without doing any computations, explain why the student's claim must be incorrect.
 (b) Find all points on the ellipse $x^2 - xy + y^2 = 1$ at which the tangent line is horizontal.

Exercise: 2.8

1–4 Both x and y denote functions of t that are related by the given equation. Use this equation and the given derivative information to find the specified derivative. ■

1. Equation: $y = 3x + 5$.
 - (a) Given that $dx/dt = 2$, find dy/dt when $x = 1$.
 - (b) Given that $dy/dt = -1$, find dx/dt when $x = 0$.
2. Equation: $x + 4y = 3$.
 - (a) Given that $dx/dt = 1$, find dy/dt when $x = 2$.
 - (b) Given that $dy/dt = 4$, find dx/dt when $x = 3$.
3. Equation: $4x^2 + 9y^2 = 1$.
 - (a) Given that $dx/dt = 3$, find dy/dt when $(x, y) = \left(\frac{1}{2\sqrt{2}}, \frac{1}{3\sqrt{2}}\right)$.
 - (b) Given that $dy/dt = 8$, find dx/dt when $(x, y) = \left(\frac{1}{3}, -\frac{\sqrt{2}}{9}\right)$.
4. Equation: $x^2 + y^2 = 2x + 4y$.
 - (a) Given that $dx/dt = -5$, find dy/dt when $(x, y) = (3, 1)$.
 - (b) Given that $dy/dt = 6$, find dx/dt when $(x, y) = (1 + \sqrt{2}, 2 + \sqrt{3})$.

Exercise: 2.9

5–10 Confirm that the stated formula is the local linear approximation at $x_0 = 0$. ■

5. $(1+x)^{15} \approx 1 + 15x$
6. $\frac{1}{\sqrt{1-x}} \approx 1 + \frac{1}{2}x$
7. $\tan x \approx x$
8. $\frac{1}{1+x} \approx 1 - x$

33–36 Find formulas for dy and Δy . ■

33. $y = x^3$
34. $y = 8x - 4$
35. $y = x^2 - 2x + 1$
36. $y = \sin x$

45–48 Use the differential dy to approximate Δy when x changes as indicated. ■

45. $y = \sqrt{3x-2}$; from $x = 2$ to $x = 2.03$
46. $y = \sqrt{x^2+8}$; from $x = 1$ to $x = 0.97$

47. $y = \frac{x}{x^2+1}$; from $x = 2$ to $x = 1.96$
48. $y = x\sqrt{8x+1}$; from $x = 3$ to $x = 3.05$